OceanStor V500R007 MIB Interface Notes

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1 Overview

This document describes the management information bases (MIBs) of Huawei storage devices, covering the MIBs' functions, nodes, tables, and alarms.

1.1 Intended Audience

1.2 Reading Guide

1.1 Intended Audience

This document is intended for:

- Application program development personnel

1.2 Reading Guide

This document gives the MIB overview and multiple MIB reference documents:

Chapter	Content
1. MIB Introduction	Network management, SNMP-based network management, Simple Network Management Protocol (SNMP), and MIB concepts and operations.
2. MIB References	MIB-based naming.

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2 MIB Introduction

2.1 Network Management
2.2 SNMP-based Network Management
2.3 SNMP Introduction
2.4 MIB Introduction
2.5 MIB Browser Introduction
2.6 Query Return Value Description

2.1 Network Management

The ever expanding network and increasing network devices present a major challenge in managing networks effectively and providing high-quality network services. Network management is vital to the proper functionality of a network solution.

Network management has four elements:

- Managed node: The device that is to be managed on the network.
- Agent: The software or hardware that tracks managed nodes.
- Manager: Communicates with the Agents in different managed nodes and displays the status of the managed nodes.
- Network management protocol: Enables information exchange between the Manager and the Agents.

Currently, the most widely adopted network management protocol for TCP/IP network is SNMP.

2.2 SNMP-based Network Management

The SNMP-based network management involves four parts:

- Network Management Station (NMS)

  The NMS is an independent device that runs network management programs. The network management programs provide at least one man-machine

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interface for network administrators to perform network management operations.

- SNMP Agent

  The SNMP Agent is a software module that resides on a managed device. The Agent receives and processes the packets sent by the NMS, and then sends response packets to the NMS. Under emergent conditions, the Agent sends **trap** packets proactively to communicate with the NMS.

- SNMP

  The SNMP works at the application layer of the TCP/IP network. It enables information exchange between the NMS and managed devices.

- MIB

  The MIB is a collection of managed objects. It is a bridge for communication between the NMS and the Agent. In addition, it enables standard interconnection between network management software and managed devices. Each Agent has a MIB whose object values can be abstracted or set by the NMS.

The previous four parts communicate with one another as follows:

- The NMS uses SNMP to communicate with the Agent and read or modify the MIB. In this way, the NMS monitors and manages the network devices.
- SNMP enables communication between the NMS and the Agent. The Protocol Data Unit (PDU) of SNMP implements information exchange. The SNMP does not handle data transfer by itself, but User Datagram Protocol (UDP) does instead.
- The Agent communicates with the NMS to maintain the MIB, to monitor and manage each module of the device.
- The MIB stores information about each module of the device. The NMS accesses the information to monitor and manage the device.

2.3 SNMP Introduction

2.3.1 Versions

SNMP has three versions: SNMPv1, SNMPv2c, and SNMPv3.

SNMPv1 and SNMPv2c use community-based SNMP authentication. The NMS uses the community name list to control the access to devices. The Agent does not check whether the sender uses an authorized community name. In addition, SNMP messages are transmitted without encryption. As a result, SNMPv1 and SNMPv2c cannot secure authentication or privacy.

SNMPv2c is an enhanced version of SNMPv1. Compared with SNMPv1, SNMPv2c supports more operations, data types, communications protocols, and error handling codes.

SNMPv3 provides all functions of SNMPv1 and SNMPv2c, and offers a more advanced security mechanism for authentication and encryption.

SNMPv3 ensures data security and access control.

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SNMPv3 enables message-level data security in terms of the three aspects:

- Data integrity

  Data modification must be authorized beforehand. Data sequence modification must be within the specific scope.
- Data origin authentication

  This authentication makes clear which user the received data comes from. SNMPv3 uses user-based security authentication. It verifies a user who sends messages, but not an application program that generates messages.
- Data verification check

  The NMS or Agent checks the time a received message was generated. The message will not be accepted if the difference between the message time and the system time exceeds the specified range. Data verification check protects in-transfer messages against being tampered with, and prevents the reception and handling of malicious messages.

  SNMPv3 implements access control based on protocol operations, and controls the access to managed objects.

2.3.2 Packet Handling

The Agent receives request packets from the NMS through UDP port 161.

The Agent processes a received packet as follows:

1. Decodes the packet. According to ASN.1 encoding rules, the Agent generates a packet that is represented in the internal data structure. If decoding fails due to an expected error, the Agent discards the packet.
2. Checks the SNMP version. The Agent compares the version of the received packet with the list of supported versions. If the version is excluded from the list, the Agent discards the packet.
3. Authenticates the community name. The Agent obtains the community name from the packet. The community name is filled by the NMS sending this packet request. If the community name is excluded from the list of supported community names, the Agent discards the packet and generates a **trap** packet.
4. Abstracts the PDU. The Agent abstracts the PDU from the successfully authenticated ASN.1 object. If the abstraction fails, the Agent discards the packet.
5. Handles the PDU. The Agent handles the PDU according to its SNMP entity to learn the MIB tree node of the managed **variable**. Then the Agent obtains the value of the managed **variable** from the corresponding module to generate a response packet. After that, the Agent encodes the response packet and sends it to the NMS.
6. The NMS receives the response packet and handles it in a similar way. Then the NMS displays a result.

2.4 MIB Introduction

The MIB is a collection of managed objects. It defines attributes of managed objects. The attributes are:

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- Name of the object
- Permission for the object
- Data type of the object

The Structure of Management Information (SMI) specifies the definition and organization of each managed object. In addition, it specifies the types of data applicable to the MIB, such as **Counter** and **Gauge**.

The MIB designates network element variables, specifically, information that can be queried and configured by the NMS. The MIB gives a data structure for the collection of possible managed objects.

2.5 MIB Browser Introduction

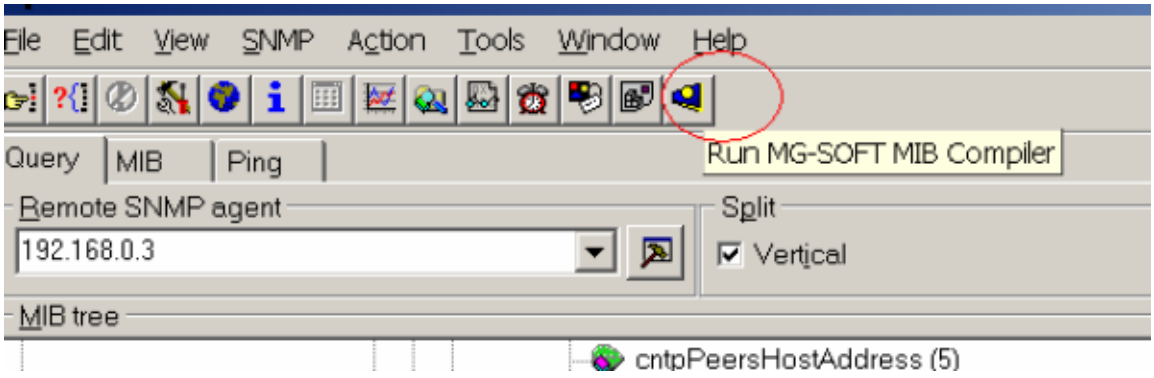
When you load MIB files, load the ones to be imported first.

Here we use the MG-SOFT MIB Browser Professional SNMPv3 Edition as an example to explain how to load MIB files.

2.5.1 Compiling and Loading MIB Files

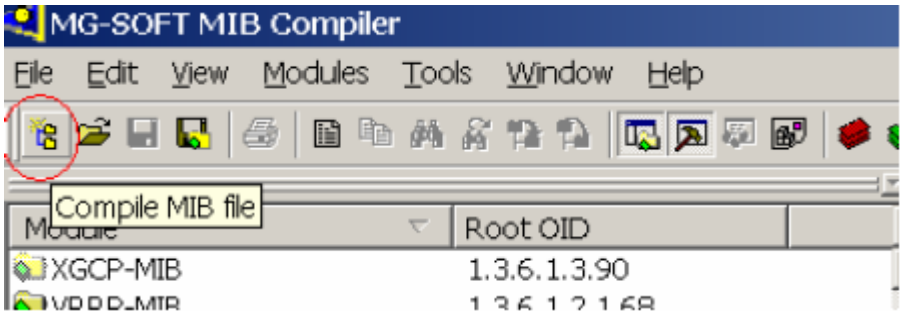
Step 1 Start the MG-SOFT MIB browser. On the **MIB Browser** window, click the **Run MG-SOFT MIB Compiler** icon, as shown in **Figure 2-1**:

Figure 2-1 MIB browser window



Step 2 On the **MG-SOFT MIB Compiler** window that is displayed, click the **Compile MIB file** icon, as shown in **Figure 2-2**:

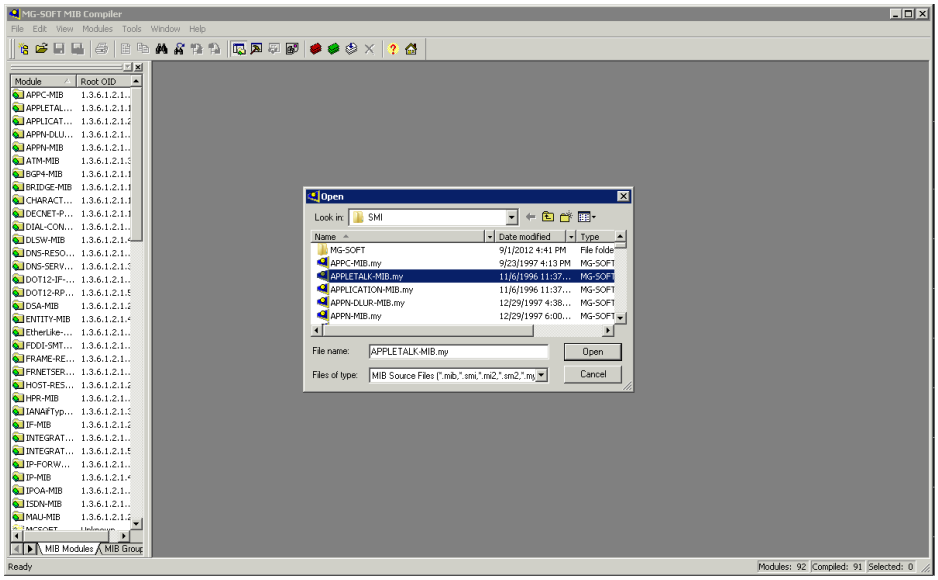
Figure 2-2 Selecting MIB compiling



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Step 3 Select the MIB file that you want to compile.

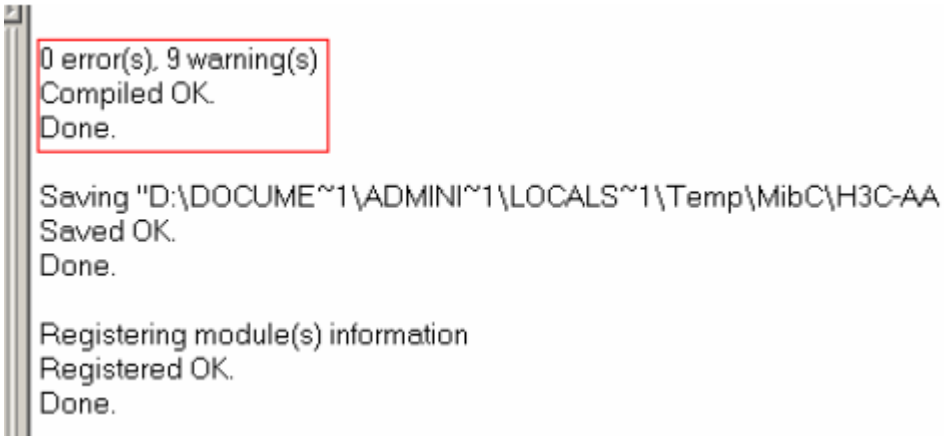
Figure 2-3 Selecting a MIB file



Step 4 Compile the selected MIB file.

Select the MIB file that you want to compile and click **Open**. If the compile succeeds, the following information is displayed:

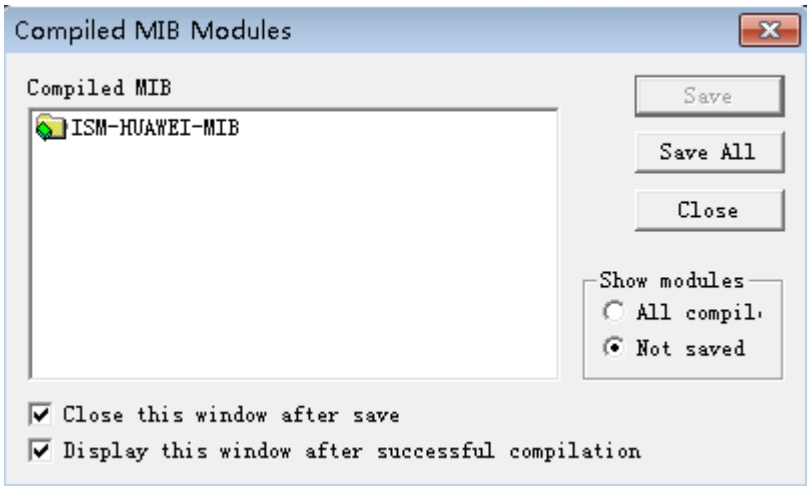
Figure 2-4 Information indicating a successful compile



Step 5 Save the successfully compiled MIB file. The MIB file will be saved in the default folder.

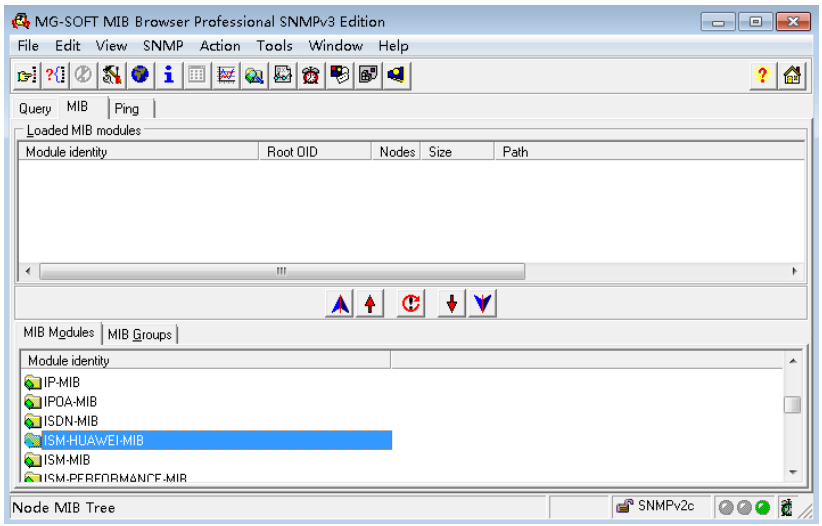
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Figure 2-5 Saving the MIB file



Step 6 Select the MIB file **ISM-HUAWEI-MIB** in **MIB/MIB Modules** column, click the red arrow to upload MIB file into the MIB Browser.

----End



2.5.2 Configuring SNMP protocol parameters

Timeout: Indicates the timeout interval of request packets. The value ranges from 1s to 30s. 15s is recommended.

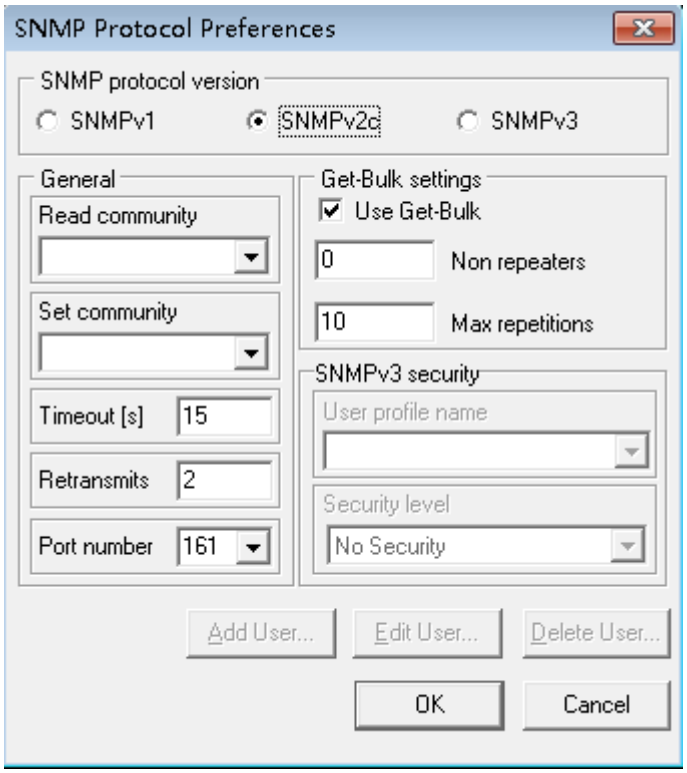
Retransmits: Indicates the resending times of request packets after the request packets timed out. The value ranges from 0 to 10. 2 is recommended. You can set the value based on the network environment.

Non repeaters: Indicates that the Get Next operation is executed only once for the first **Non repeaters** parameters in binding parameters. If the parameter value is greater than the total number of binding parameters, the total number of binding parameters prevails. If the parameter value is smaller than 0, **0** prevails. You are advised to set the parameter value to **0**. Parameter settings vary depending on site requirements.

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Max repetitions: Indicates the maximum number of repeated Get Next operations performed for all parameters except the first **Non repeaters** parameters in binding parameters. The value ranges from 1 to 50. 10 is recommended. A too large value results in **tooBig** errors.

Note: Parameters **Non repeaters** and **Max repetitions** are unavailable for the SNMPv1 protocol.



2.6 Query Return Value Description

There are two forms of snmp query return value:

One is OID + index value + data:

```
33: hwStorageFCPortID.1122304 (Unsigned32) 1122304
34: hwStorageFCPortID.1122305 (Unsigned32) 1122305
35: hwStorageFCPortID.1122306 (Unsigned32) 1122306
36: hwStorageFCPortID.1122307 (Unsigned32) 1122307
```

The other is OID + index value length + index value + data:

```
43: hwPerfDiskIndex.1.48 (OCTET STRING) 0 [
44: hwPerfDiskIndex.1.49 (OCTET STRING) 1 [
45: hwPerfDiskIndex.1.50 (OCTET STRING) 2 [
46: hwPerfDiskIndex.1.51 (OCTET STRING) 3 [
```

The index value is generated internally by us as an index of data for querying. For example, when querying hard disk data, there are multiple hard disks in the system. The internal query will be OID + index value query to avoid data confusion on multiple disks.

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The index value is the value obtained by the first OID in each table. The type of the index value is the cause of the difference in the return value.

1. The type of the index value is a string, then the presentation form is: ascll code length + ascll code value of the index value (because the type is a string, its length is required to continue the query).

2. The type of the index value is non-string, so the presentation form is: index value (non-string, the index value can continue to query)

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3 RFC MIB Support

3.1 RFC 1213 mib-2 System Module

3.1 RFC 1213 mib-2 System Module

sysDescr

Node	Data Type	Meaning	Permission	Status
sysDescr OID: 1.3.6.1.2.1.1.1	OCTET STRING	System description	Read-only	Current

sysObjectID

Node	Data Type	Meaning	Permission	Status
sysObjectID OID: 1.3.6.1.2.1.1.2	Object Identifier	System OID	Read-only	Current

sysUpTime

Node	Data Type	Meaning	Permission	Status
sysUpTime OID: 1.3.6.1.2.1.1.3	Timeticks	Running time of the SNMP service. Unit: 1% second.	Read-only	Current

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sysName

Node	Data Type	Meaning	Permission	Status
sysName OID: 1.3.6.1.2.1.1.5	OCTET STRING	System name	Read-write	Current

sysLocation

Node	Data Type	Meaning	Permission	Status
sysLocation OID: 1.3.6.1.2.1.1.6	OCTET STRING	System location	Read-write	Current

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4 MIB References for Huawei Nodes (Standard Enterprise Node: 2011)

4.1 Common Module
4.2 Alarm Module
4.3 Alarm Module(alarm:251)
4.4 Storage Hardware Modules
4.5 TOPO Module
4.6 SAN Module

4.1 Common Module

patch version

Node	Data Type	Meaning	Permission	Status
patchversion OID: 1.3.6.1.4.1.2011.2.251.1.1.7	OCTET STRING	Patch version.	Read-only	Current

4.2 Alarm Module

hwIsmActiveAlarmInfoTable

This table saves active alarms, which can be queried by the NMS.

The indexes to this table are **1: hwIsmActiveAlarmInfoNodeCode** and **2: hwIsmActiveAlarmInfoSerialNo**.

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The OID of this table is 1.3.6.1.4.1.2011.2.91.10.1.1.

Node	Data Type	Meaning	Permission	Status
hwIsmActiveAlarmInfoNodeCode OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.1	OCTET STRING	ID of the node where an alarm occurs.	Read-only	Current
hwIsmActiveAlarmInfoLocationInfo OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	Read-only	Current
hwIsmActiveAlarmInfoRestoreAdvice OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.3	OCTET STRING	Alarm handling suggestion.	Read-only	Current
hwIsmActiveAlarmInfoTitle OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.4	OCTET STRING	Alarm information title.	Read-only	Current
hwIsmActiveAlarmInfoType OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.5	Integer	Alarm type: 2-EquipmentAlarm	Read-only	Current
hwIsmActiveAlarmInfoLevel OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.6	Integer	Alarm severity of I2000 network management software: 1. -criticalAlarm 2. -majorAlarm 3. -minorAlarm 4. -warningAlarm	Read-only	Current
hwIsmActiveAlarmInfoAlarmID OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.7	Gauge	Alarm ID of I2000 network management software.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwIsmActiveAlarmInfoOccurTime OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.8	OCTET STRING	Time when an alarm was generated.	Read-only	Current
hwIsmActiveAlarmInfoSerialNo OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.9	Gauge	Alarm serial number (SN)	Read-only	Current
hwIsmActiveAlarmInfoAddtionInfo OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.10	OCTET STRING	Alarm cause	Read-only	Current
hwIsmActiveAlarmInfoCategory OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.11	Integer	Alarm category: 1. -faultAlarm 2. -resumeAlarm 3. -eventAlarm	Read-only	Current
hwIsmActiveAlarmInfoLocalAlarmID OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.12	Counter 64	Original alarm ID	Read-only	Current

hwIsmTrapTargetAddrTable

This table gives information about alarm forwarding addresses, involving IP addresses, ports, and versions.

The index to this table is **hwIsmTrapTargetAddrIndex**.

The OID of this table is 1.3.6.1.4.1.2011.2.91.10.4.1.

Node	Data Type	Meaning	Permission	Status

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hwIsmTrapTargetAddrIPAddr OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.1	IpAddress	Destination address for trap forwarding. Only IP addresses in the IPv4 format are supported. If domain names are supported and valid, the destination address is the IP address corresponding to the domain name; if the domain name is invalid, "0.0.0.0" is displayed.	Read-create	Current
hwIsmTrapTargetAddrPort OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.2	Integer	Port of the destination address for trap forwarding. The value can be 1 to 65535.	Read-create	Current
hwIsmTrapTargetAddrRowStatus OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.3	Integer	Row status. The value can be: ● 1: Modify the Trap forwarding destination address ● 4: Add the Trap forwarding destination address ● 6: Delete the Trap forwarding destination address	Read-create	Current
hwIsmTrapTargetAddrIndex OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.4	OCTET STRING	Index of the destination address for trap forwarding. The value can be 0 to 3.	Read-create	Current
hwIsmTrapTargetAddrTrapVer OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.5	Integer	Version of the destination address for trap forwarding. The value can be: ● 1: SNMPv1 ● 2: SNMPv2c ● 3: SNMPv3 When configuring the destination address for trap forwarding, if this field is not filled, the default is SNMPv2c.	Read-create	Current

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hwIsmTrapTargetAddrI-PAddrNew OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.6	OCTET STRING	New destination address for trap forwarding. IP addresses in the IPv4 or IPv6 format, and IP address in the domain name format are supported. If **hwIsmTrapTargetAddrI-PAddr** and **hwIsmTrapTargetAddrI-PAddrNew** fields are filled with values, take the value of the **hwIsmTrapTargetAddrI-PAddrNew** field by default. The domain name address is not configured on the DNS server or the network delays, leading to the timeout of the message returned by the network management software. Adjust the timeout period setting based on the network environment.	Read-create	Current

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hwIsmTrapTargetAddr-TrapType OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.7	Integer	Type of the destination address for trap forwarding. The value can be: ● 1: Parsed. Sends parsed alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.2011.2.91.10.2.1.0.1) ● 2: Original. Sends original alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.34774.4.1.4.2). ● 3: All. sends Parsed, Original, and Parsed_alarm_oid alarms to the destination address for trap forwarding. ● 4: Parsed_alarm_oid. sends parsed alarms to the destination address for trap forwarding. (Each alarm corresponds to one OID. Products later than V3R6 support the alarm OID). ● 5: Parsed_time_string. Sends parsed alarms to the destination address for trap forwarding (Products V3R3C20SPC200 and later than V3R6 support this trap type. Trap OID: 1.3.6.1.4.1.2011.2.251.20.1.2.1) ● 6: Original_time_string. Sends original alarms to the destination address for trap forwarding (Products V3R3C20SPC200 and later than V3R6 support this trap type. Trap OID:1.3.6.1.4.1.2011.2.251.20.2.1.1).	Read-create	Current

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		When configuring the destination address for trap forwarding, if this field is not filled, the default is Parsed.		
hwIsmTrapTargetAddrUSMUserName OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.8	OCTET STRING	USM user name of the destination address for trap forwarding. When the trap forwarding address type is SNMPv3, this field is mandatory and users must ensure a specified USM user exists in the system. Otherwise, this operation fails to be performed. If this field is mandatory, and the parameter is not delivered or no value is specified for the parameter, the system default value is **Kaimse**.	Read-create	Current

hwIsmClearedAlarmConfirm

Node	Data Type	Meaning	Permission	Status
hwIsmClearedAlarmConfirm OID: 1.3.6.1.4.1.2011.2.91.10.1.2	Gauge	Confirmation of an active alarm. **Set**: Complete the SN of an active alarm. If the SN exists, clear this alarm and a clear alarm is reported. If the SN does not exist, an error message is returned. **Get**: Returns a fixed value: 0, the value is meaningless.	Read-write	Current

hwIsmAlarmReporting

The node is used for reporting the parsed alarm.

The OID of this node is 1.3.6.1.4.1.2011.2.91.10.2.1.0.1.

Bound Variable	Data Type	Meaning	Permission	Status

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hwIsmReportingAlarmNodeCode OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.1	OCTET STRING	ID of the node where an alarm occurs.	accessible-for-notify	current
hwIsmReportingAlarmLocationInfo OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	accessible-for-notify	current
hwIsmReportingAlarmRestoreAdvice OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.3	OCTET STRING	Alarm handling suggestion.	accessible-for-notify	current
hwIsmReportingAlarmFaultTitle OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.4	OCTET STRING	Alarm information title.	accessible-for-notify	current
hwIsmReportingAlarmFaultType OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.5	Integer32	Alarm type: 2-EquipmentAlarm 3-EventAlarm	accessible-for-notify	current
hwIsmReportingAlarmFaultLevel OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.6	Integer32	Alarm severity of I2000 network management software: 1-criticalAlarm 2-majorAlarm 3-minorAlarm 4-warningAlarm 5-eventAlarm	accessible-for-notify	current
hwIsmReportingAlarmAlarmID OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.7	Gauge32	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwIsmReportingAlarmFaultTime OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.8	DateAndTime	Time when an alarm was generated.	accessible-for-notify	current
hwIsmReportingAlarmSerialNo OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.9	Gauge32	Alarm serial number (SN)	accessible-for-notify	current

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hwIsmReportingAlarmAdditionInfo OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.10	OCTET STRING	Alarm cause	accessible-for-notify	current
hwIsmReportingAlarmFaultCategory OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.11	Integer32	Alarm category: 1-faultAlarm 2-resumeAlarm 3-eventAlarm	accessible-for-notify	current
hwIsmReportingAlarmLocationAlarmID OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.12	Counter64	Original alarm ID	accessible-for-notify	current
hwIsmReportingAlarmProductModel OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.13	Integer32	Product Model	accessible-for-notify	current
hwIsmReportingAlarmProductSN OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.14	OCTET STRING	SN	accessible-for-notify	current
hwIsmReportingAlarmProductSystemName OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.15	OCTET STRING	System name	accessible-for-notify	current

4.3 Alarm Module(alarm:251)

hwStorageActiveAlarmInfoTable

This table saves active alarms, which can be queried by the NMS.

The indexes to this table are **1: hwStorageActiveAlarmInfoNodeCode** and **2: hwStorageActiveAlarmInfoSerialNo**.

The OID of this table is 1.3.6.1.4.1.2011.2.251.20.1.1.1.

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Node	Data Type	Meaning	Permission	Status
hwStorageActiveAlarmInfoNodeCode OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.1	OCTET STRING	ID of the node where an alarm occurs.	Read-only	Current
hwStorageActiveAlarmInfoLocationInfo OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	Read-only	Current
hwStorageActiveAlarmInfoRestoreAdvice OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.3	OCTET STRING	Alarm handling suggestion.	Read-only	Current
hwStorageActiveAlarmInfoTitle OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.4	OCTET STRING	Alarm information title.	Read-only	Current
hwStorageActiveAlarmInfoType OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.5	Integer	Alarm type: 2-EquipmentAlarm	Read-only	Current
hwStorageActiveAlarmInfoLevel OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.6	Integer	Alarm severity of I2000 network management software: 1. -criticalAlarm 2. -majorAlarm 3. -minorAlarm 4. -warningAlarm	Read-only	Current
hwStorageActiveAlarmInfoAlarmID OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.7	Gauge	Alarm ID of I2000 network management software.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageActiveAlarmInfoOccurTime OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.8	OCTET STRING	Time when an alarm was generated.	Read-only	Current
hwStorageActiveAlarmInfoSerialNo OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.9	Gauge	Alarm serial number (SN)	Read-only	Current
hwStorageActiveAlarmInfoAddtionInfo OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.10	OCTET STRING	Alarm cause	Read-only	Current
hwStorageActiveAlarmInfoCategory OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.11	Integer	Alarm category: 1. -faultAlarm 2. -resumeAlarm 3. -eventAlarm	Read-only	Current
hwStorageActiveAlarmInfoLocalAlarmID OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.12	Counter64	Original alarm ID	Read-only	Current

hwStorageAlarmReporting

The node is used for reporting the parsed alarm.

The OID of this node is 1.3.6.1.4.1.2011.2.251.20.1.2.1.

Bound Variable	Data Type	Meaning	Permission	Status
hwStorageReportingAlarmNodeCode OID:1.3.6.1.4.1.2011.2.251.20.1.3.1	OCTET STRING	ID of the node where an alarm occurs.	accessible-for-notify	current

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hwStorageReportingAlarm-LocationInfo OID:1.3.6.1.4.1.2011.2.251.20.1.3.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	accessible-for-notify	current
hwStorageReportingAlarm-RestoreAdvice OID:1.3.6.1.4.1.2011.2.251.20.1.3.3	OCTET STRING	Alarm handling suggestion.	accessible-for-notify	current
hwStorageReportingAlarm-FaultTitle OID:1.3.6.1.4.1.2011.2.251.20.1.3.4	OCTET STRING	Alarm information title.	accessible-for-notify	current
hwStorageReportingAlarm-FaultType OID:1.3.6.1.4.1.2011.2.251.20.1.3.5	Integer32	Alarm type: 2-EquipmentAlarm 3-EventAlarm	accessible-for-notify	current
hwStorageReportingAlarm-FaultLevel OID:1.3.6.1.4.1.2011.2.251.20.1.3.6	Integer32	Alarm severity of I2000 network management software: 1-criticalAlarm 2-majorAlarm 3-minorAlarm 4-warningAlarm 5-eventAlarm	accessible-for-notify	current
hwStorageReportingAlarmAlarmID OID:1.3.6.1.4.1.2011.2.251.20.1.3.7	Gauge32	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwStorageReportingAlarm-FaultTime OID:1.3.6.1.4.1.2011.2.251.20.1.3.8	OCTET STRING	Time when an alarm was generated.	accessible-for-notify	current
hwStorageReportingAlarm-SerialNo OID:1.3.6.1.4.1.2011.2.251.20.1.3.9	Gauge32	Alarm serial number (SN)	accessible-for-notify	current
hwStorageReportingAlarmAdditionInfo OID:1.3.6.1.4.1.2011.2.251.20.1.3.10	OCTET STRING	Alarm cause	accessible-for-notify	current

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hwStorageReportingAlarm-FaultCategory OID:1.3.6.1.4.1.2011.2.251.20.1.3.11	Integer32	Alarm category: 1-faultAlarm 2-resumeAlarm 3-eventAlarm	accessible-for-notify	current
hwStorageReportingAlarm-LocationAlarmID OID:1.3.6.1.4.1.2011.2.251.20.1.3.12	Counter64	Original alarm ID	accessible-for-notify	current
hwStorageReportingAlarm-ProductModel OID:1.3.6.1.4.1.2011.2.251.20.1.3.13	Integer32	Product Model	accessible-for-notify	current
hwStorageReportingAlarm-ProductSN OID:1.3.6.1.4.1.2011.2.251.20.1.3.14	OCTET STRING	SN	accessible-for-notify	current
hwStorageReportingAlarm-ProductSystemName OID:1.3.6.1.4.1.2011.2.251.20.1.3.15	OCTET STRING	System name	accessible-for-notify	current

hwStorageEventType

This table is used exclusively for ISM management software and is not available to others.

The OID of this node is 1.3.6.1.4.1.2011.2.251.20.2.1.1.

Bound Variable	Data Type	Meaning	Permission	Status
hwStorageTrapEvent-Type OID:1.3.6.1.4.1.2011.2.251.20.2.2.1	Gauge32	Alarm type. The value can be: 0: Event 1: Fault 2: Recovery	accessible-for-notify	current
hwStorageTrapEventID OID:1.3.6.1.4.1.2011.2.251.20.2.2.2	Counter64	Original alarm ID	accessible-for-notify	current

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hwStorageTrapEventLevel OID:1.3.6.1.4.1.2011.2.251.20.2.2.3	Gauge32	Alarm severity. The value can be: **2**: Info **3**: Warning **5**: Major **6**: Critical	accessible-for-notify	current
hwStorageTrapEventSequence OID:1.3.6.1.4.1.2011.2.251.20.2.2.4	Gauge32	Alarm SN, namely, unique identifier of the alarm.	accessible-for-notify	current
hwStorageTrapEventTime OID:1.3.6.1.4.1.2011.2.251.20.2.2.5	Gauge32	UTC time when the alarm was generated, unit: seconds.	accessible-for-notify	current
hwStorageTrapEventRecoveryTime OID:1.3.6.1.4.1.2011.2.251.20.2.2.6	Gauge32	UTC time when the alarm was cleared, unit: seconds.	accessible-for-notify	current
hwStorageTrapEventParameter OID:1.3.6.1.4.1.2011.2.251.20.2.2.7	OCTET STRING	Alarm location parameter. This parameter can be used to resolve alarm description. It is used with the alarm configuration file of the upper-layer network management system.	accessible-for-notify	current
hwStorageTrapEventID32Bit OID:1.3.6.1.4.1.2011.2.251.20.2.2.8	Gauge32	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwStorageTrapEventTimeStr OID:1.3.6.1.4.1.2011.2.251.20.2.2.9	OCTET STRING	Time when an alarm was generated. string format.	accessible-for-notify	current
hwStorageTrapEventRecoveryTimeStr OID:1.3.6.1.4.1.2011.2.251.20.2.2.10	OCTET STRING	Time when an alarm was cleared. string format.	accessible-for-notify	current

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4.4 Storage Hardware Modules

hwInfoPortIBTable

This table is used to query information about InfiniBand ports.

The index to this table is **hwInfoPortIBID**.

The OID of this table is 1.3.6.1.4.1.2011.2.251.22.16500. Value "4294967295" or all-F indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortIBID OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.1	OCTET STRING	ID of an InfiniBand port, table index	read-only	current
hwInfoPortIBParentType OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.2	Gauge	Parent type of an InfiniBand port. Possible values are: **207**: controller **208**: expansion board **209**: interface module	read-only	current
hwInfoPortIBParentID OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.3	OCTET STRING	Parent ID of an InfiniBand port	read-only	current
hwInfoPortIBLocation OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.4	OCTET STRING	Location of an InfiniBand port	read-only	current
hwInfoPortIBHealthStatus OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.5	Gauge	Health status of an InfiniBand port. For details, see **HEALTH_STATUS_E**.	read-only	current

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hwInfoPortIBRunningStatus OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.6	Gauge	Operating status of an InfiniBand port. For details, see **RUNNING_STATUS_E**.	read-only	current
hwInfoPortIBType OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.7	Gauge	Logical type of an InfiniBand port. For details, see **PORT_LOGIC_TYPE_E**.	read-only	current
hwInfoPortIBWorkingRate OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.8	Gauge	Working rate of an InfiniBand port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	read-only	current
hwInfoPortIBWWN OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.9	OCTET STRING	WWN of an InfiniBand port	read-only	current
hwInfoPortIBRole OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.10	Gauge	Role of an InfiniBand port. Possible values are: **2**: initiator **3**: target **4**: initiator and target **4294967295**: invalid	read-only	current
hwInfoPortIBSymbolError OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.11	Gauge	Symbol error statistics of an InfiniBand port	read-only	current
hwInfoPortIBLinkErrorRecovery OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.12	Gauge	Statistics of rectifying an InfiniBand port's connection errors	read-only	current

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hwInfoPortIBReceiveErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.13	Gauge	InfiniBand port receiving errors	read-only	current
hwInfoPortIBRemoteReceiveErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.14	Gauge	InfiniBand port remotely receiving errors	read-only	current
hwInfoPortIBReceiveTransmitErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.15	Gauge	InfiniBand port receiving and forwarding errors	read-only	current
hwInfoPortIBNotSendPacakges OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.16	Gauge	Count of packets that are not sent by an InfiniBand port	read-only	current
hwInfoPortIBReceiveConstraintErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.17	Gauge	InfiniBand port receiving constraint errors	read-only	current
hwInfoPortIBLinkErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.18	Gauge	Connection error of an InfiniBand port	read-only	current
hwInfoPortIBBufferOverrunErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.19	Gauge	Buffer overflow error of an InfiniBand port	read-only	current
hwInfoPortIBStartTime OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.20	OCTET STRING	Start time of an InfiniBand port	read-only	current
hwInfoPortIBMaxSpeed OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.21	Gauge	Maximum rate of an InfiniBand port (unit: Mbit/s)	read-only	current

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4.5 TOPO Module

hwIsmAccessNodeTable

This table gives basis device information.

The index to this table is **hwIsmNENodeCode**.

The OID of this table is 1.3.6.1.4.1.2011.2.91.9.1.1.

Node	Data Type	Meaning	Permission	Status
hwIsmNENodeCode OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.1	OCTET STRING	ID of an NE node.	Read-only	Current
hwIsmNENodeType OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.2	Integer	Type of an NE node.	Read-only	Current
hwIsmNENodeWorkingMode OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.3	Integer	Working mode of an NE node, **not used currently.**	Read-only	Current
hwIsmNENodeIPAddress OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.4	IpAddress	IP address of an NE node, **not used currently.**	Read-only	Current
hwIsmNENodeContextName OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.5	OCTET STRING	Name of the context.	Read-only	Current
hwIsmNENodeContextEngineID OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.6	OCTET STRING	Engine ID of the context, **not used currently.**	Read-only	Current

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hwIsmNENodeCluster-Name OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.7	OCTET STRING	Name of the NE cluster, **not used currently.**	Read-only	Current
hwIsmNENodeRunning-Status OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.8	Integer	Running status of an NE: 1. The SNMP agent communicates correctly with the device. 2. The SNMP agent fails to communicate with the device. **Not used currently.**	Read-only	Current

4.6 SAN Module

hwInfoMappingViewTable

This table provides information about MappingViews.

The index to this table is **hwInfoMappingViewID**.

The OID of this table is 1.3.6.1.4.1.2011.2.251.23.245.

Node	Data Type	Meaning	Permission	Status
hwInfoMappingViewID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.1	OCTET STRING	ID of an Mapping View.	Read-only	Current
hwInfoMappingView-Name OID: 1.3.6.1.4.1.2011.2.251.23.245.1.2	OCTET STRING	Name of Mapping View.	Read-only	Current
hwInfoMappingViewIn-bandCommand OID: 1.3.6.1.4.1.2011.2.251.23.245.1.3	Gauge	Whether to enable the command device 0: enable 1: disable	Read-only	Current

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hwInfoMappingView-LunGroupID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.4	OCTET STRING	ID of Lun Group in Mapping View.	Read-only	Current
hwInfoMappingView-LunGroupName OID: 1.3.6.1.4.1.2011.2.251.23.245.1.5	OCTET STRING	Name of Lun Group in Mapping View.	Read-only	Current
hwInfoMappingViewHostGroupID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.6	OCTET STRING	ID of Host Group in Mapping View.	Read-only	Current
hwInfoMappingViewHostGroupName OID: 1.3.6.1.4.1.2011.2.251.23.245.1.7	OCTET STRING	Name of Lun Group in Mapping View.	Read-only	Current
hwInfoMappingView-PortGroupID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.8	OCTET STRING	ID of Port Group in Mapping View.	Read-only	Current
hwInfoMappingView-PortGroupName OID: 1.3.6.1.4.1.2011.2.251.23.245.1.9	OCTET STRING	Name of Port Group in Mapping View.	Read-only	Current

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5 MIB References for Storage Nodes (Standard Enterprise Node: 34774)

5.1 Common Module
5.2 Alarm Module
5.3 Performance Statistics Module
5.4 Physical Storage Modules
5.5 Logical Storage Modules
5.6 Storage Space Modules
5.7 Storage Hardware Modules
5.8 NAS Information Query Module

5.1 Common Module

deviceId

Node	Data Type	Meaning	Permission	Status
deviceId OID: 1.3.6.1.4.1.34774.4.1.1.1	OCTET STRING	Device serial number, namely, unique identifier of the device.	Read-only	Current

deviceType

Node	Data Type	Meaning	Permission	Status

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deviceType OID: 1.3.6.1.4.1.34774.4.1.1.2	Integer	Type of the storage device. For details, see **PRODUCT_MODE_E**.	Read-only	Current

status

Node	Data Type	Meaning	Permission	Status
status OID: 1.3.6.1.4.1.34774.4.1.1.3	Integer	System operating status. For details, see **RUNNING_STATUS_E**.	Read-only	Current

usedCapacity

Node	Data Type	Meaning	Permission	Status
usedCapacity OID: 1.3.6.1.4.1.34774.4.1.1.4	Counter64	Used capacity of the storage device. The unit is MB.	Read-only	Current

totalCapacity

Node	Data Type	Meaning	Permission	Status
totalCapacity OID: 1.3.6.1.4.1.34774.4.1.1.5	Counter64	Total capacity of the storage device. The unit is MB.	Read-only	Current

version

Node	Data Type	Meaning	Permission	Status

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version OID: 1.3.6.1.4.1.34774.4.1.1.6	OCTET STRING	Device version.	Read-only	Current

deviceModel

Node	Data Type	Meaning	Permission	Status
deviceModel OID: 1.3.6.1.4.1.34774.4.1.1.7	OCTET STRING	Model of the storage device.	Read-only	Current

5.2 Alarm Module

eventTable

This table gives device alarm information, which can be queried. This table is used exclusively for ISM management software and is not available to others.

The index to this table is **hwIsmEventSequence**.

The OID of this node is 1.3.6.1.4.1.34774.4.1.3.1.

Node	Data Type	Meaning	Permission	Status
hwIsmEventType OID: 1.3.6.1.4.1.34774.4.1.3.1.1.1	Gauge	Alarm type. The value can be: **0**: eventAlarm **1**: faultAlarm **2**: resumeAlarm	Read-write	Current
hwIsmEventID OID: 1.3.6.1.4.1.34774.4.1.3.1.1.2	Counter64	ID of an alarm. It indicates a collection of alarms with the same category.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwIsmEventLevel OID: 1.3.6.1.4.1.34774.4.1.3.1.1.3	Gauge	Alarm severity. The value can be: **2**: Info **3**: Warning **5**: Major **6**: Critical	Read-write	Current
hwIsmEventSequence OID: 1.3.6.1.4.1.34774.4.1.3.1.1.4	Gauge	Alarm SN, namely, unique identifier of the alarm.	Read-write	Current
hwIsmEventTime OID: 1.3.6.1.4.1.34774.4.1.3.1.1.5	Gauge	UTC time when the alarm was generated. The unit is second.	Read-write	Current
hwIsmEventRecoveryTime OID: 1.3.6.1.4.1.34774.4.1.3.1.1.6	Gauge	UTC time when the alarm was cleared. The unit is second.	Read-write	Current
hwIsmEventParameter OID: 1.3.6.1.4.1.34774.4.1.3.1.1.7	OCTET STRING	Alarm location parameter. This parameter can be used to resolve alarm description. It is used with the alarm configuration file of the upper-layer network management system.	Read-write	Current
hwIsmEventRowStatus OID: 1.3.6.1.4.1.34774.4.1.3.1.1.20	Integer	Row status value. The value can be: 1: Row Active 6: Remove a alarm according to the alarm serial number.	Read-write	Current

eventType

This table is used exclusively for ISM management software and is not available to others.

The OID of this node is 1.3.6.1.4.1.34774.4.1.4.2.

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Bound Variable	Data Type	Meaning	Permission	Status
hwIsmTrapEventType OID:1.3.6.1.4.1.34774.4.1.3.3.1	Gauge	Alarm type. The value can be: 0: Event 1: Fault 2: Recovery	accessible-for-notify	current
hwIsmTrapEventID OID:1.3.6.1.4.1.34774.4.1.3.3.2	Counter64	Original alarm ID	accessible-for-notify	current
hwIsmTrapEventLevel OID:1.3.6.1.4.1.34774.4.1.3.3.3	Gauge	Alarm severity. The value can be: **2**: Info **3**: Warning **5**: Major **6**: Critical	accessible-for-notify	current
hwIsmTrapEventSequence OID:1.3.6.1.4.1.34774.4.1.3.3.4	Gauge	Alarm SN, namely, unique identifier of the alarm.	accessible-for-notify	current
hwIsmTrapEventTime OID:1.3.6.1.4.1.34774.4.1.3.3.5	Gauge	UTC time when the alarm was generated, unit: seconds.	accessible-for-notify	current
hwIsmTrapEventRecoveryTime OID:1.3.6.1.4.1.34774.4.1.3.3.6	Gauge	UTC time when the alarm was cleared, unit: seconds.	accessible-for-notify	current
hwIsmTrapEventParameter OID:1.3.6.1.4.1.34774.4.1.3.3.7	OCTET STRING	Alarm location parameter. This parameter can be used to resolve alarm description. It is used with the alarm configuration file of the upper-layer network management system.	accessible-for-notify	current
hwIsmTrapEventID32Bit OID:1.3.6.1.4.1.34774.4.1.3.3.8	Gauge	Alarm ID of I2000 network management software.	accessible-for-notify	current

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hwIsmTrapEventTimeStr OID:1.3.6.1.4.1.34774.4.1.3.3.9	OCTET STRING	Time when an alarm was generated. string format.	accessible-for-notify	current
hwIsmTrapEventRecoveryTimeStr OID:1.3.6.1.4.1.34774.4.1.3.3.10	OCTET STRING	Time when an alarm was cleared. string format.	accessible-for-notify	current

forwardAddrTable

This table gives information about alarm forwarding address information, involving IP addresses, ports, and versions. This table is used exclusively for ISM management software and is not available to others.

The index to this table is **forwardAddrIndex**.

The OID of this node is 1.3.6.1.4.1.34774.4.1.2.1.

Node	Data Type	Meaning	Permission	Status
forwardAddrIndex OID: 1.3.6.1.4.1.34774.4.1.2.1.1.1	OCTET STRING	Index to the alarm forwarding address. The value can be 0 to 3.	Read-write	Current
forwardAddrIP OID: 1.3.6.1.4.1.34774.4.1.2.1.1.2	IpAddress	IP address to which the alarm is forwarded. Only support IPv4 format. If domain names are supported and valid, the destination address is the IP address corresponding to the domain name; if the domain name is invalid, "0.0.0.0" is displayed.	Read-write	Current
forwardAddrPort OID:1 .3.6.1.4.1.34774.4.1.2.1.1.3	Gauge	Port of the alarm forwarding address. The value can be 1 to 65535.	Read-create	Current

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forwardAddrTrapVer OID: 1.3.6.1.4.1.34774.4.1.2.1.1.4	Integer	Version of the alarm forwarding address. The value can be: • 1: SNMPv1 • 2: SNMPv2c • 3: SNMPv3 When configuring the alarm forwarding address, if this field is not filled, the default is SNMPv2c.	Read-create	Current
forwardAddrRowStatus OID:1 .3.6.1.4.1.34774.4.1.2.1.1.5	Integer	Row status value. The value can be: • 1: Modify the alarm forwarding address • 4: Add the alarm forwarding address • 6: Delete the alarm forwarding address	Read-create	Current
forwardAddrIPNew OID: 1.3.6.1.4.1.34774.4.1.2.1.1.6	OCTET STRING	New alarm forwarding address. IP addresses in the IPv4 or IPv6 format, and IP address in the domain name format are supported.. If **forwardAddrIP** and **forwardAddrIPNew** fields are filled with values, take the value of the **forwardAddrIPNew** field by default. The domain name address is not configured on the DNS server or the network delays, leading to the timeout of the message returned by the network management software. Adjust the timeout period setting based on the network environment.	Read-create	Current

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forwardAddrTrapType OID: 1.3.6.1.4.1.34774.4.1.2.1.1.7	Integer	Type of the alarm forwarding address. The value can be: ● 1: Parsed. Sends parsed alarms to the alarm forwarding address (Trap OID:1.3.6.1.4.1.2011.2.91.10.2.1.0.1) ● 2: Original. Sends original alarms to the alarm forwarding address (Trap OID:1.3.6.1.4.1.34774.4.1.4.2). ● 3: All. sends Parsed, Original, and Parsed_alarm_oid alarms to the destination address for trap forwarding. ● 4: Parsed_alarm_oid. sends parsed alarms to the destination address for trap forwarding. (Each alarm corresponds to one OID. Products later than V3R6 support the alarm OID). ● 5: Parsed_time_string. Sends parsed alarms to the destination address for trap forwarding (Products V3R3C20SPC200 and later than V3R6 support this trap type. Trap OID:1.3.6.1.4.1.2011.2.251.20.1.2.1) ● 6: Original_time_string. Sends original alarms to the destination address for trap forwarding (Products V3R3C20SPC200 and later than V3R6 support this trap type. Trap OID:1.3.6.1.4.1.2011.2.251.20.2.1.1). When configuring the alarm forwarding address, if this field is not filled, the default is Original.		

forwardAddrUSMU-serName OID: 1.3.6.1.4.1.34774.4.1.2.1.1.8	OCTET STRING	USM user name of the destination address for trap forwarding. When the trap forwarding address type is SNMPv3, this field is mandatory and users must ensure a specified USM user exists in the system. Otherwise, this operation fails to be performed. If this field is mandatory, and the parameter is not delivered or no value is specified for the parameter, the system default value is **Kaimse**.	Read-create	Current

5.3 Performance Statistics Module

Note: Due to the limitation of the snmp protocol, it is not suitable for querying data of specific large-size logical objects, and it is not suitable for performance statistics.

Recommendation: Use devicemanager / CLI to query performance statistics.

hwPerformanceSwitch

This node indicates the status of the performance statistics switch.

Node	Data Type	Meaning	Permission	Status
hwPerformanceSwitch OID: 1.3.6.1.4.1.34774.4.1.21.1	Integer	Performance statistics switch. The value can be: 0: Disabled 1: Enabled	Read-write	Current

hwPerfNodeTable

This table is used to query the performance data of nodes.

The index to this table is **hwPerfNodeIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.3. The value "4294967295" or "18446744073709551615"or "F" of performance statistics entry indicates unknown or invalid.

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Node	Data Type	Meaning	Permission	Status
hwPerfNodeIfIndex OID: 1.3.6.1.4.1.34774.4.1.21.3.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwPerfNodeCPUUsage OID: 1.3.6.1.4.1.34774.4.1.21.3.1.2	Gauge	CPU usage of the node (unit: %).	Read-only	Current
hwPerfNodeAvgCacheUsage OID: 1.3.6.1.4.1.34774.4.1.21.3.1.3	Gauge	Average cache usage of the node (unit: %).	Read-only	Current
hwPerfNodeDelay OID: 1.3.6.1.4.1.34774.4.1.21.3.1.4	Gauge	Average I/O latency of the node (unit: ms).	Read-only	Current
hwPerfNodeTotalIOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.5	Gauge	Total IOPS of the node (unit: IO/s).	Read-only	Current
hwPerfNodeReadIOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.6	Gauge	Read IOPS of the node (unit: IO/s).	Read-only	Current
hwPerfNodeWriteIOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.7	Gauge	Write IOPS of the node (unit: IO/s).	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwPerfNodeTotalTraffic OID: 1.3.6.1.4.1.34774.4.1.21.3.1.8	Gauge	Total bandwidth for the node (unit: MB/s).	Read-only	Current
hwPerfNodeReadTraffic OID: 1.3.6.1.4.1.34774.4.1.21.3.1.9	Gauge	Read bandwidth for the node (unit: MB/s).	Read-only	Current
hwPerfNodeWriteTraffic OID: 1.3.6.1.4.1.34774.4.1.21.3.1.10	Gauge	Write bandwidth for the node (unit: MB/s).	Read-only	Current
hwPerfNodeFileBandwidth OID: 1.3.6.1.4.1.34774.4.1.21.3.1.11	Counter64	Node file bandwidth (unit: byte/s)	Read-only	Current
hwPerfNodeFileOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.12	Counter64	Node file OPS	Read-only	Current

hwPerfPortTable

This table is used to query the performance data of ETH, FC, FCoE, Bond ports.

The index to this table is **hwPerfPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.5. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwPerfPortIfIndex OID: 1.3.6.1.4.1.34774.4.1.21.5.1.1	Gauge	Index to the port.	Read-only	Current
hwPerfPortDelay OID: 1.3.6.1.4.1.34774.4.1.21.5.1.2	Gauge	I/O latency of the port (unit: ms).	Read-only	Current

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hwPerfPortTotalIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.3	Gauge	IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortReadIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.4	Gauge	Read IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortWriteIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.5	Gauge	Write IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortTotalTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.6	Gauge	Total bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortReadTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.7	Gauge	Read bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortWriteTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.8	Gauge	Write bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortMaxTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.9	Gauge	Maximum bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortMaxIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.10	Gauge	Maximum IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortLocation OID: 1.3.6.1.4.1.34774.4.1.21.5.1.11	OCTET STRING	Location of the port.	Read-only	Current

hwPerfLunTable

This table is used to query the performance data (skip the grayed texts) of LUNs.

The index to this table is **hwPerfLunID**.

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The OID of this table is 1.3.6.1.4.1.34774.4.1.21.4. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwPerfLunID OID:1.3.6.1.4.1.34774.4.1.21.4.1.1	Gauge	LUN ID.	Read-only	Current
hwPerfLunHitRate OID:1.3.6.1.4.1.34774.4.1.21.4.1.2	Gauge	I/O hit ratio of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunTotalIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.3	Gauge	Current IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunReadIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.4	Gauge	Current read IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunWriteIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.5	Gauge	Current write IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunTotalTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.6	Gauge	Current total bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunReadTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.7	Gauge	Current read bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunWriteTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.8	Gauge	Current write bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunReadIORate OID:1.3.6.1.4.1.34774.4.1.21.4.1.9	Gauge	Read I/O ratio of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunWriteIORate OID:1.3.6.1.4.1.34774.4.1.21.4.1.10	Gauge	Write I/O ratio of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunMaxTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.11	Gauge	Maximum bandwidth for the LUN (unit: MB/s).	Read-only	Current

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hwPerfLunMaxIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.12	Gauge	Maximum IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunAverageIOResponseTime OID:1.3.6.1.4.1.34774.4.1.21.4.1.13	Gauge	Average IO response time (unit:us).	Read-only	Current
hwPerfLunMaxIOResponseTime OID:1.3.6.1.4.1.34774.4.1.21.4.1.14	Gauge	Maximum IO response time (unit:us).	Read-only	Current
hwPerfLunAverageReadIOLatency OID:1.3.6.1.4.1.34774.4.1.21.4.1.15	Gauge	Average read IO response time (unit:us).	Read-only	Current
hwPerfLunAverageWriteIOLatency OID:1.3.6.1.4.1.34774.4.1.21.4.1.16	Gauge	Average write IO response time (unit:us).	Read-only	Current

hwPerfCacheTable

This table is used to query the performance data of caches.

The index to this table is **hwPerfCacheID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.7. The value of performance statistics entry is "-1", indicates invalid.

Node	Data Type	Meaning	Permission	Status

hwPerfCacheID OID:1.3.6.1.4.1.34774.4.1.21.7.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwPerfCacheReadUtilization OID:1.3.6.1.4.1.34774.4.1.21.7.1.2	Integer	Cache read usage (unit: %).	Read-only	Current
hwPerfCacheWriteUtilization OID:1.3.6.1.4.1.34774.4.1.21.7.1.3	Integer	Cache write usage (unit: %).	Read-only	Current
hwPerfCacheMirrorWriteUtilization OID:1.3.6.1.4.1.34774.4.1.21.7.1.4	Integer	Cache mirror write usage (unit: %).	Read-only	Current
hwPerfCacheHitRatio OID:1.3.6.1.4.1.34774.4.1.21.7.1.5	Integer	Cache hit ratio.	Read-only	Current

hwPerfControllerNFSV3Table

This table is used to query the NFSv3 performance data of controllers.

The index to this table is **hwPerfControllerNFSV3ID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.10. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

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Node	Data Type	Description	Permission	Status
hwPerfControllerNFSV3ID OID: 1.3.6.1.4.1.34774.4.1.21.10.1.1	OCTET STRING	Controller ID.	Not-accessible	Current
hwPerfControllerNFSV3OPS OID: 1.3.6.1.4.1.34774.4.1.21.10.1.2	Counter64	NFSv3 OPS of the controller	Read-only	Current
hwPerfControllerNFSV3Bps OID: 1.3.6.1.4.1.34774.4.1.21.10.1.3	Counter64	NFSv3 throughput of the controller (unit: byte/s)	Read-only	Current

hwPerfControllerNFSV4Table

This table is used to query the NFSv4 performance data of controllers.

The index to this table is **hwPerfControllerNFSV4ID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.11. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwPerfControllerNFSV4ID OID: 1.3.6.1.4.1.34774.4.1.21.11.1.1	OCTET STRING	Controller ID.	Not-accessible	Current
hwPerfControllerNFSV4OPS OID: 1.3.6.1.4.1.34774.4.1.21.11.1.2	Counter64	NFSv4 OPS of the controller	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwPerfControllerNFSV4Bps OID: 1.3.6.1.4.1.34774.4.1.21.11.1.3	Counter64	NFSv4 throughput of the controller (unit: byte/s)	Read-only	Current

hwPerfControllerNFSV41Table

This table is used to query the NFSv41 performance data of controllers.

The index to this table is **hwPerfControllerNFSV41ID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.15. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwPerfControllerNFSV4ID OID: 1.3.6.1.4.1.34774.4.1.21.15.1.1	OCTET STRING	Controller ID.	Not-accessible	Current
hwPerfControllerNFSV4OPS OID: 1.3.6.1.4.1.34774.4.1.21.15.1.2	Counter64	NFSv4 OPS of the controller	Read-only	Current
hwPerfControllerNFSV4Bps OID: 1.3.6.1.4.1.34774.4.1.21.15.1.3	Counter64	NFSv4 throughput of the controller (unit: byte/s)	Read-only	Current

hwPerfControllerSMBV1Table

This table is used to query the SMB V1 performance data of controllers.

The index to this table is **hwPerfControllerSMBV1ID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.12. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

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Node	Data Type	Description	Permission	Status
hwPerfControllerSMBV1ID OID: 1.3.6.1.4.1.34774.4.1.21.12.1.1	OCTET STRING	Controller ID	Not-accessible	Current
hwPerfControllerSMBV1OPS OID: 1.3.6.1.4.1.34774.4.1.21.12.1.2	Counter64	SMB V1 OPS of the controller	Read-only	Current
hwPerfControllerSMBV1Bps OID: 1.3.6.1.4.1.34774.4.1.21.12.1.3	Counter64	SMB V1 throughput of the controller (unit: byte/s)	Read-only	Current

hwPerfControllerSMBV2Table

This table is used to query the SMB V2 performance data of controllers.

The index to this table is **hwPerfControllerSMBV2ID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.13. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwPerfControllerSMBV2ID OID: 1.3.6.1.4.1.34774.4.1.21.13.1.1	OCTET STRING	Controller ID	Not-accessible	Current
hwPerfControllerSMBV2OPS OID: 1.3.6.1.4.1.34774.4.1.21.13.1.2	Counter64	SMB V2 OPS of the controller	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwPerfControllerSMBV2Bps OID: 1.3.6.1.4.1.34774.4.1.21.13.1.3	Counter64	SMB V2 throughput of the controller (unit: byte/s)	Read-only	Current

hwPerfFileSystemTable

This table is used to query the performance data of a file system.

The index to this table is hwPerfFileSystemID.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.14. -1 indicates that the performance statistics are invalid.

Node	Data Type	Description	Permission	Status
hwPerfFileSystemID OID: 1.3.6.1.4.1.34774.4.1.21.14.1.1	OCTET STRING	File system ID	Not-accessible	Current
hwPerfFileSystemOPS OID: 1.3.6.1.4.1.34774.4.1.21.14.1.2	Counter64	File system OPS	Read-only	Current
hwPerfFileSystemServiceTime OID: 1.3.6.1.4.1.34774.4.1.21.14.1.3	Counter64	Service time of the file system (unit: μs)	Read-only	Current
hwPerfFileSystemReadOPS OID: 1.3.6.1.4.1.34774.4.1.21.14.1.4	Counter64	Read OPS of the file system	Read-only	Current
hwPerfFileSystemReadBandwidth OID: 1.3.6.1.4.1.34774.4.1.21.14.1.5	Counter64	Read bandwidth of the file system (unit: MB/s)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwPerfFileSystemAvRdOPSRspTime OID: 1.3.6.1.4.1.34774.4.1.21.14.1.6	Counter64	Average read OPS response time of the file system (unit: μs)	Read-only	Current
hwPerfFileSystemAvWrOPSRspTime OID: 1.3.6.1.4.1.34774.4.1.21.14.1.7	Counter64	Average write OPS response time of the file system (unit: μs)	Read-only	Current
hwPerfFileSystemWriteOPS OID: 1.3.6.1.4.1.34774.4.1.21.14.1.8	Counter64	Write OPS of the file system	Read-only	Current
hwPerfFileSystemWriteBandwidth OID: 1.3.6.1.4.1.34774.4.1.21.14.1.9	Counter64	Write bandwidth of the file system (unit: MB/s)	Read-only	Current

hwPerfDiskTable

This table is used to query the performance of a disk.

The index to this table is hwPerfDiskIndex.

The OID of this table is 1.3.6.1.4.1.2011.2.251.21.10. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwPerfDiskIndex OID: 1.3.6.1.4.1.2011.2.251.21.10.1.1	OCTET STRING	Disk ID.	read-only	current
hwPerfDiskTotalIOPS OID: 1.3.6.1.4.1.2011.2.251.21.10.1.2	Gauge	IOPS (unit: IO/s).	read-only	current

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hwPerfDiskReadIOPS OID: 1.3.6.1.4.1.2011.2.251.21.10.1.3	Gauge	Read IOPS (unit: IO/s).	read-only	current
hwPerfDiskWriteIOPS OID: 1.3.6.1.4.1.2011.2.251.21.10.1.4	Gauge	Write IOPS (unit: IO/s).	read-only	current
hwPerfDiskTotalTraffic OID: 1.3.6.1.4.1.2011.2.251.21.10.1.5	Counter64	Bandwidth (unit: MB/s) .	read-only	current
hwPerfDiskReadTraffic OID: 1.3.6.1.4.1.2011.2.251.21.10.1.6	Counter64	Read bandwidth (unit: MB/s) .	read-only	current
hwPerfDiskWriteTraffic OID: 1.3.6.1.4.1.2011.2.251.21.10.1.7	Counter64	Write bandwidth (unit: MB/s) .	read-only	current
hwPerfDiskDelay OID: 1.3.6.1.4.1.2011.2.251.21.10.1.8	Counter64	Response time (unit: μs) .	read-only	current
hwPerfDiskLengthOfQueue OID: 1.3.6.1.4.1.2011.2.251.21.10.1.9	Gauge	Queue length (unit:-).	read-only	current
hwPerfDiskAverageIO OID: 1.3.6.1.4.1.2011.2.251.21.10.1.11	Gauge	Average I/O size (unit: KB).	read-only	current
hwPerfDiskAverageReadIO OID: 1.3.6.1.4.1.2011.2.251.21.10.1.12	Gauge	Average read I/O size (unit: KB).	read-only	current
hwPerfDiskAverageWriteIO OID: 1.3.6.1.4.1.2011.2.251.21.10.1.13	Gauge	Average write I/O size (unit: KB).	read-only	current

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hwPerfDiskMaxIOPS OID: 1.3.6.1.4.1.2011.2.251.21.10.1.14	Gauge	Maximum IOPS (unit: IO/s).	read-only	current

5.4 Physical Storage Modules

Note: If the value of an interface parameter is 4294967295 or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.

hwStorageFCPortTable

This table provides information about FC ports.

The index to this table is **hwStorageFCPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.7.

Node	Data Type	Meaning	Permission	Status
hwStorageFCPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.1	Gauge	Index to the FC host port.	Read-only	Current
hwStorageFCPortBoardIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.2	Gauge	Index to the board of the FC host port.	Read-only	Current
hwStorageFCPortID OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.3	Gauge	ID of the FC host port.	Read-only	Current
hwStorageFCPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.4	Integer	Status of the FC host port. The value can be: **0**: Disconnected **1**: Connected **2**: Faulty	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageFCPortConfigRate OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.5	Integer	Working Rate of the FC host port, (unit: 1000Mbit/s). The value can be: **0**: Auto **1**: GBPS-1 **2**: GBPS-2 **4**: GBPS-4 **8**: GBPS-8 **16**: GBPS-16 The Current rate can be **1 Gbit/s**, **2 Gbit/s**, **4 Gbit/s**, **8 Gbit/s**, or **16 Gbit/s**. The conversion unit is 1000. **0** indicates auto-negotiation. **-1** indicates invalid.	Read-only	Current
hwStorageFCPortMode OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.6	Integer	Mode of the FC host port. The value can be: ● -1: Invalid ● 0: fabric ● 1: FC-AL ● 2: P2P ● 3: Auto	Read-only	Current
hwStorageFCPortWWN OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.7	OCTET STRING	WWN of the FC host port.	Read-only	Current
hwStorageFCPortLogicType OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.8	Integer	Logical type of FC host port. The value can be: ● 1: service ● 2: ibc ● 3: ibs ● 4: expansion ● 5: management ● 6: upsmanagement ● 8: maintenance	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageFCPortSpeedRate OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.9	Integer	Speed rate of FC host port(unit: Mbit/s). If the parameter value is **-1**, this parameter is invalid.	Read-only	Current

hwStorageSASPortTable

This table provides information about SAS ports.

The index to this table is **hwStorageSASPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.8.

Node	Data Type	Meaning	Permission	Status
hwStorageSASPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.1	Gauge	Index to the SAS host port.	Read-only	Current
hwStorageSASPortBoardIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.2	Gauge	Index to the board of the SAS host port.	Read-only	Current
hwStorageSASPortID OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.3	Gauge	ID of the SAS host port.	Read-only	Current
hwStorageSASPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.4	Integer	Status of the SAS host port. The value can be: **0**: disconnected **1**: connected **2**: faulty	Read-only	Current
hwStorageSASPortConfigRate OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.5	Integer	Rate of the SAS host port. (unit: 1000 Mbit/s). If the parameter value is **-1**, this parameter is invalid.	Read-only	Current
hwStorageSASPortWWN OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.6	OCTET STRING	WWN of the SAS host port.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageSASPortLogicType OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.7	Integer	Logical type of SAShost port. The value can be: • 1: service • 2: ibc • 3: ibs • 4: expansion • 5: management • 6: upsmanagement • 8: maintenance	Read-only	Current
hwStorageSASPortSpeedRate OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.8	Integer	Speed rate of SAS host port. The value can be: • 1: Gbps15 • 2: Gbps30 • 3: Gbps60	Read-only	Current

hwStorageISCSIPortTable

This table provides information about iSCSI ports.

The index to this table is **hwStorageISCSIPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.9.

Node	Data Type	Meaning	Permission	Status
hwStorageISCSIPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.1	Gauge	Index to the iSCSI host port.	Read-only	Current
hwStorageISCSIPortBoardIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.2	Gauge	Index to the board of the iSCSI host port.	Read-only	Current
hwStorageISCSIPortID OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.3	Gauge	ID of the iSCSI host port.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageISCSIPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.4	Integer	Status of the iSCSI host port. The value can be: 0: disconnected 1: connected 2: faulty	Read-only	Current
hwStorageISCSIPortIP OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.5	IpAddress	IP address of the iSCSI host port.	Read-only	Current
hwStorageISCSIPortNetMask OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.6	IpAddress	Mask of the iSCSI host port.	Read-only	Current
hwStorageISCSIPortBindMode OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.7	Integer	Bind mode of the iSCSI host port, **not used currently**.	Read-only	Current
hwStorageISCSIPortLogicType OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.8	Integer	Logical type of ISCSI host port. The value can be: ● 1: service ● 2: ibc ● 3: ibs ● 4: expansion ● 5: management ● 6: upsmanagement ● 8: maintenance	Read-only	Current

hwStorageFrontEndHostPortTable

This table provides basic information about the SAS, FC, iSCSI ports whose performance statistics can be collected.

The index to this table is **hwStorageFrontEndHostPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.11.

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Node	Data Type	Meaning	Permission	Status
hwStorageFrontEndHostPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.1	Gauge	Combination index to the host ports.	Read-only	Current
hwStorageFrontEndHostPortType OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.2	Integer	Physical type of a host port: • 0: SAS • 1: FC • 5: iSCSI	Read-only	Current
hwStorageFrontEndHostPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.3	Integer	Connection state of a host port: • 0: link down • 1: link up • 2: not exist • 3: link bad • 4: hardware error • 5: speed no match • 6: enable • 7: disable • 8: mac error • 9: lane no match	Read-only	Current
hwStorageFrontEndHostPortPhysAddress OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.4	OCTET STRING	IP address of a host port. If "--" is displayed, the value is invalid.	Read-only	Current
hwStorageFrontEndHostPortDescription OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.5	OCTET STRING	Description of a host port.	Read-only	Current

hwStorageControllerTable

This table provides basic information about the controller.

The index to this table is **hwStorageControllerID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.12.

Node	Data Type	Meaning	Permission	Status

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hwStorageControllerID OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwStorageController-Name OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.2	OCTET STRING	Controller name	Read-only	Current
hwStorageControllerLocation OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.3	OCTET STRING	Controller location	Read-only	Current
hwStorageController-HealthStatus OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.4	Integer	Controller health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwStorageControllerRunningStatus OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.5	Integer	Controller running status. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwStorageController-SoftVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.6	OCTET STRING	Controller software version	Read-only	Current
hwStorageController-Temperature OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.7	Integer	Controller temperature, unit: degree, not used currently.	Read-only	Current

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hwStorageControllerIsMaster OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.8	Gauge	The state of controller, master or slave 0: Slave 1: Master Not used currently.	Read-only	Current
hwStorageControllerELabel OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.9	OCTET STRING	Controller electronic label	Read-only	Current
hwStorageControllerPCBVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.10	OCTET STRING	Controller PCB version	Read-only	Current
hwStorageControllerBMCVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.11	OCTET STRING	Controller BMC version. When the value is empty or "--", indicates unknown or invalid.	Read-only	Current
hwStorageControllerLogicVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.12	OCTET STRING	Controller logical version	Read-only	Current
hwStorageControllerBIOSVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.13	OCTET STRING	Controller BIOS version	Read-only	Current
hwStorageControllerMemorySize OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.14	Gauge	Controller memory size, unit: MB	Read-only	Current
hwStorageControllerCPUInfo OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.15	OCTET STRING	CPU information, composed by four fields: manufacturer, make, model, and frequency.	Read-only	Current
hwStorageControllerVoltage OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.16	Gauge	Controller voltage, unit: 0.1 V.	Read-only	Current

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5.5 Logical Storage Modules

*Note: If the value of an interface parameter is **4294967295** or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.*

hwStorageLunTable

This table is used to query the LUN information about a device.

The index to this table is **hwStorageLunID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.9.4.

Node	Data Type	Meaning	Permission	Status
hwStorageLunID OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.1	Gauge	ID of the LUN.	Read-only	Current
hwStorageLunName OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.2	OCTET STRING	Name of the LUN.	Read-only	Current
hwStorageLunWWN OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.3	OCTET STRING	WWN of the LUN.	Read-only	Current
hwStorageLunPoolID OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.4	Gauge	ID of the storage pool to which the LUN belongs.	Read-only	Current
hwStorageLunCapacity OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.5	Counter 64	Capacity of the LUN. The unit is KB.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageLunOwning-Controller OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.6	Gauge	Node id of the owning controller of the LUN. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwStorageLunStripDepth OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.7	Gauge	Stripe depth of the LUN. **Not used currently.**	Read-only	Current
hwStorageLunWriteS-trategy OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.8	Integer	Write policy of the LUN. The value can be: ● 1: Write through ● 2: Write back no mirroring ● 3: Write back mirroring ● 4: Write back mandatory nomirroring ● 5: Write back mandatory mirroring	Read-only	Current
hwStorageLunPrefetch-Strategy OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.9	Integer	Prefetch policy of the LUN. The value can be: ● 0: Non-prefetch ● 1: Constant prefetch ● 2: Variable prefetch ● 3: Intelligent prefetch	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageLunPrefetch-Value OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.10	Gauge	Cache prefetch value of the LUN. The cache prefech policy can be: **1**. constant prefetch (unit: KB) **2**. variable prefetch (unit: times) **3**. non-prefetch or intelligent prefetch. If **4294967295** is returned, the value is invalid.	Read-only	Current
hwStorageLunStatus OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.11	Integer	Status of the LUN. The value can be: • 1: Normal • 2: Faulty • 3: Unformatted • 4: Formatting	Read-only	Current
hwStorageLunRowStatus OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.12	Integer	Row status. • 1: Active	Read-write	Current

hwStorageCacheTable

This table provides cache information such as cache ID, cache size, cache high water level, and cache low water level.

The index to this table is **hwStorageCacheID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.9.1.

Node	Data Type	Meaning	Permission	Status

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hwStorageCacheID OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwStorageCacheTotal-Capacity OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.2	Gauge	Cache size (unit: MB).	Read-only	Current
hwStorageCacheHigh-WaterLevel OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.3	Gauge	Cache high water level.	Read-only	Current
hwStorageCacheLow-WaterLevel OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.4	Gauge	Cache low water level.	Read-only	Current
hwStorageCacheRow-Status OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.5	Integer	Row status. ● 1: Active	Read-write	Current

5.6 Storage Space Modules

Note: If the value of an interface parameter is 4294967295 or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.

hwInfoDiskDomainTable

This table is used to check the disk domain information about a device. The index to this table is **hwInfoDiskDomainID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.1.

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskDomainID OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.1	OCTET STRING	Disk ID, table index	Read-only	Current
hwInfoDiskDomainName OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.2	OCTET STRING	Name of a disk domain	Read-only	Current
hwInfoDiskDomainHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.3	Gauge	Health status of a disk domain. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoDiskDomainRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.4	Gauge	Running status of a disk domain. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoDiskDomainTotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.5	Counter64	Total capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainFreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.6	Counter64	Free capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.7	Counter64	Hot spare capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainUsedHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.8	Counter64	Used hot spare capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier0DiskNumber OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.9	Gauge	Total number of disks on tier 0 of a disk domain. (unit: block)	Read-only	Current
hwInfoDiskDomainTier0TotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.10	Counter64	Total capacity of tier 0 of a disk domain. (unit: MB)	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskDomainTier0FreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.11	Counter64	Free capacity of tier 0 of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier0HotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.12	Counter64	Total hot spare capacity of tier 0 of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier0UsedHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.13	Counter64	Used hot spare capacity of tier 0 of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier0HotSpareStrategy OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.14	Counter64	Hot spare policy of tier 0 of a disk domain. Possible values are: • 0: None • 1: Low • 2: High • 3: None	Read-only	Current
hwInfoDiskDomainTier1DiskNumber OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.15	Gauge	Total number of disks on tier 1 of a disk domain. (unit: block)	Read-only	Current
hwInfoDiskDomainTier1TotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.16	Gauge	Total capacity of tier 1 of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier1FreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.17	Counter64	Free capacity of tier 1 of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier1HotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.18	Counter64	Total hot spare capacity of tier 1 of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainTier1UsedHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.19	Counter64	Used hot spare capacity of tier 1 of a disk domain. (unit: MB)	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskDomainTier1HotSpareStrategy OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.20	Gauge	Hot spare policy of tier 1 of a disk domain. Possible values are: ● 0: None ● 1: Low ● 2: High ● 3: None	Read-only	Current
hwInfoDiskDomainTier2DiskNumber OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.21	Gauge	Total number of disks on tier 2 of a disk domain (unit: block)	Read-only	Current
hwInfoDiskDomainTier2TotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.22	Counter64	Total capacity of tier 2 of a disk domain (unit: MB)	Read-only	Current
hwInfoDiskDomainTier2FreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.23	Counter64	Free capacity of tier 2 of a disk domain (unit: MB)	Read-only	Current
hwInfoDiskDomainTier2HotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.24	Counter64	Total hot spare capacity of tier 2 of a disk domain (unit: MB)	Read-only	Current
hwInfoDiskDomainTier2UsedHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.25	Counter64	Used hot spare capacity of tier 2 of a disk domain (unit: MB)	Read-only	Current
hwInfoDiskDomainTier2HotSpareStrategy OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.26	Gauge	Hot spare policy of tier 2 of a disk domain. Possible values are: ● 0: None ● 1: Low ● 2: High ● 3: None	Read-only	Current

hwInfoStoragePoolTable

This table is used to check the storage pool information about a device. The index to this table is **hwInfoStoragePoolID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.2.

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Node	Data Type	Description	Permission	Status
hwInfoStoragePoolID OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.1	OCTET STRING	ID of a storage pool, table index	Read-only	Current
hwInfoStoragePoolName OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.2	OCTET STRING	Name of a storage pool	Read-only	Current
hwInfoStoragePoolDiskDomainID OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.3	OCTET STRING	ID of the disk domain where a storage pool resides	Read-only	Current
hwInfoStoragePoolDiskDomainName OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.4	OCTET STRING	Name of the disk domain where a storage pool resides	Read-only	Current
hwInfoStoragePoolHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.5	Gauge	Health status of a storage pool. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoStoragePoolRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.6	Gauge	Operating status of a storage pool. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoStoragePoolTotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.7	Counter64	Total capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolSubscribedCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.8	Counter64	Used capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolFreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.9	Counter64	Available capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolProtectionCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.10	Counter64	Data protection capacity of a storage pool (unit: MB)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoStoragePoolTier0Capacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.11	Counter64	Tier 0 raw capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolTier1Capacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.12	Counter64	Tier 1 raw capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolTier2Capacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.13	Counter64	Tier 2 raw capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolFullThreshold OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.14	Gauge	Threshold of used capacity of a storage pool (unit: %)	Read-only	Current
hwInfoStoragePoolExtentSize OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.15	Gauge	Relocation granularity of a storage pool (unit: KB)	Read-only	Current
hwInfoStoragePoolSmartTierFeatureStatus OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.16	Gauge	Dynamic storage tiering (DST) status of a storage pool. Possible values are: ● 1: Active ● 2: Inactive	Read-only	Current
hwInfoStoragePoolRelocationStatus OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.17	Gauge	Relocation status of a storage pool. Possible values are: ● 1: Ready ● 2: Migrating ● 3: Paused	Read-only	Current
hwInfoStoragePoolRelocationTriggerMode OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.18	Gauge	Relocation triggering mode of a storage pool. Possible values are: ● 1: Schedule ● 2: Manual	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoStoragePoolRelocationPaused OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.19	Gauge	To determine the relocation of a storage pool is paused. Possible values are: • 0: No • 1: Yes	Read-only	Current
hwInfoStoragePoolEstimatedMoveUpData OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.20	Counter64	Amount of data to be promoted in a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolEstimatedMoveDownData OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.21	Counter64	Amount of data to be demoted in a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolEstimatedDataRelocationDuration OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.22	Counter64	Relocation duration of a storage pool (unit: second)	Read-only	Current
hwInfoStoragePoolLunTotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.23	Counter64	Lun capacity of a storage pool (unit: MB)	read-only	current
hwInfoStoragePoolDataReductionRatio OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.24	OCTET STRING	Data Reduction Ratio	read-only	current
hwInfoStoragePoolDeduplicationRatio OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.25	OCTET STRING	Data Deduplication Ratio	read-only	current
hwInfoStoragePoolCompressionRatio OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.26	OCTET STRING	Data Compression Ratio	read-only	current

hwInfoStorageTierTable

This table is used to check the storage tier information about a device. The index to this table is **hwInfoStorageTierID**.

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The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.3.

Node	Data Type	Description	Permission	Status
hwInfoStorageTierID OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.1	OCTET STRING	ID of a storage tier, table index, the format of storage tier ID is: tier 0: PoolID.0 tier 1: PoolID.1 tier 2: PoolID.2	Read-only	Current
hwInfoStorageTierName OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.2	OCTET STRING	Name of a storage tier	Read-only	Current
hwInfoStorageTierPoolID OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.3	OCTET STRING	ID of the storage pool where a storage tier resides	Read-only	Current
hwInfoStorageTier-HealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.4	Gauge	Health status of a storage tier. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoStorageTierRun-ningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.5	Gauge	Operating status of a storage tier. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoStorageTierCapac-ity OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.6	Counter64	Total capacity of a storage tier (unit: MB)	Read-only	Current
hwInfoStorageTierAllo-catedCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.7	Counter64	Used capacity of a storage tier (unit: MB)	Read-only	Current
hwInfoStorageTierFree-Capacity OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.8	Counter64	Available capacity of a storage tier (unit: MB)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoStorageTierRAIDLevel OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.9	Gauge	RAID level of a storage tier. Possible values are: • 0: None • 1: RAID 10 • 2: RAID 5 • 3: RAID 0 • 4: RAID 1 • 5: RAID 6 • 6: RAID 50 • 7: RAID 3 • 8: RAID 60 • 9: EC1 • 10: EC2 • 11: EC3	Read-only	Current
hwInfoStorageTierRAIDDiskNumber OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.10	Gauge	Number of member disks in a RAID group of a storage tier	Read-only	Current
hwInfoStorageTierEstimatedMoveUpData OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.11	Counter64	Amount of data to be promoted on a storage tier (unit: MB)	Read-only	Current
hwInfoStorageTierEstimatedMoveDownData OID:1.3.6.1.4.1.34774.4.1.23.4.3.1.12	Counter64	Amount of data to be demoted on a storage tier (unit: MB)	Read-only	Current

hwInfoPortGroupTable

This table is used to check the port group information about a device. The index to this table is **hwInfoPortGroupID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.4.

Node	Data Type	Description	Permission	Status
hwInfoPortGroupID OID:1.3.6.1.4.1.34774.4.1.23.4.4.1.1	OCTET STRING	ID of a port group, table index	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortGroupName OID:1.3.6.1.4.1.34774.4.1.23.4.4.1.2	OCTET STRING	Name of a port group	Read-only	Current
hwInfoPortGroupPortList OID:1.3.6.1.4.1.34774.4.1.23.4.4.1.3	OCTET STRING	List of ports in a port group. The format of list is: ID1: Type1: Location1, ID2: Type2: Location2…. The value of Type is: • 213: Port ETH • 212: Port FC • 252: Port FCoE	Read-only	Current

hwInfoHostTable

This table is used to check the host information about a device. The index to this table is **hwInfoHostID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.5.

Node	Data Type	Description	Permission	Status
hwInfoHostID OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.1	OCTET STRING	Host ID, table index	Read-only	Current
hwInfoHostName OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.2	OCTET STRING	Host name	Read-only	Current
hwInfoHostLocation OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.3	OCTET STRING	Host location	Read-only	Current
hwInfoHostHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.4	Gauge	Host health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoHostRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.5	Gauge	Host operating status. For details, see **RUNNING_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoHostOperating-System OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.6	Gauge	Host operating system. Possible values are: • 0: Linux • 1: Windows • 2: Solaris • 3: HP-UX • 4: AIX • 5: XenServer • 6: Mac OS • 7: VMware ESX • 8: VIS6000 • 9: Windows Server 2012 • 10: Oracle_VM • 11: OpenVMS • 12: Oracle_VM_Server_for_x86 • 13: Oracle_VM_Server_for_SPARC	Read-only	Current
hwInfoHostIPAddress OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.7	OCTET STRING	Host IP address	Read-only	Current
hwInfoHostNetwor-kName OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.8	OCTET STRING	Host domain name	Read-only	Current
hwInfoHostModel OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.9	OCTET STRING	Host model	Read-only	Current

hwInfoHostGroupTable

This table is used to check the host group information about a device. The index to this table is **hwInfoHostGroupID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.6.

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Node	Data Type	Description	Permission	Status
hwInfoHostGroupID OID:1.3.6.1.4.1.34774.4.1.23.4.6.1.1	OCTET STRING	ID of a host group, table index	Read-only	Current
hwInfoHostGroupName OID:1.3.6.1.4.1.34774.4.1.23.4.6.1.2	OCTET STRING	Name of a host group	Read-only	Current
hwInfoHostGroupHost-List OID:1.3.6.1.4.1.34774.4.1.23.4.6.1.3	OCTET STRING	List of hosts in a host group. The format of list is: ID1: Name1, ID2: Name2…	Read-only	Current

hwInfoLunGroupTable

This table is used to check the LUN group information about a device. The index to this table is **hwInfoLunGroupID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.7.

Node	Data Type	Description	Permission	Status
hwInfoLunGroupID OID:1.3.6.1.4.1.34774.4.1.23.4.7.1.1	OCTET STRING	ID of a LUN group, table index	Read-only	Current
hwInfoLunGroupName OID:1.3.6.1.4.1.34774.4.1.23.4.7.1.2	OCTET STRING	Name of a LUN group	Read-only	Current
hwInfoLunGroupLunList OID:1.3.6.1.4.1.34774.4.1.23.4.7.1.3	OCTET STRING	List of LUNs in a LUN group. The format of list is: ID1: Name1, ID2: Name2…	Read-only	Current

hwInfoLunTable

This table is used to check the LUN information about a device. The index to this table is **hwInfoLunID**.

The prefix to the OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.8.

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Node	Data Type	Description	Permission	Status
hwInfoLunID OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.1	OCTET STRING	LUN ID	Read-only	Current
hwInfoLunName OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.2	OCTET STRING	LUN name	Read-only	Current
hwInfoLunPoolID OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.3	OCTET STRING	ID of the storage pool to which the LUN belongs	Read-only	Current
hwInfoLunPoolName OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.4	OCTET STRING	Name of the storage pool to which the LUN belongs	Read-only	Current
hwInfoLunCapacity OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.5	Counter64	Total capacity of the LUN (unit: MB)	Read-only	Current
hwInfoLunSubscribedCapacity OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.6	Counter64	Capacity usage quota of the LUN (unit: MB)	Read-only	Current
hwInfoLunProtectionCapacity OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.7	Counter64	Data protection capacity quota of the LUN (unit: KB)	Read-only	Current
hwInfoLunSectorSize OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.8	Gauge	Sector size of the LUN (unit: B)	Read-only	Current
hwInfoLunHealthStatus OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.9	Gauge	Health status of the LUN. For details, see **HEALTH_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLunRunningStatus OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.10	Gauge	Operating status of the LUN. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoLunType OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.11	Gauge	LUN type. The value can be: **0**: thick LUN **1**: thin LUN	Read-only	Current
hwInfoLunIOPriority OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.12	Gauge	I/O priority of the LUN. The value can be: **1**: low **2**: middle **3**: high	Read-only	Current
hwInfoLunWWN OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.13	OCTET STRING	WWN of the LUN	Read-only	Current
hwInfoLunExposedToInitiator OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.14	Gauge	Whether the LUN is mapped to an initiator. The value can be: **0**: no **1**: yes	Read-only	Current
hwInfoLunWritePolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.15	Gauge	Cache write policy of the LUN. The value can be: **1**: write back **2**: write through **3**: forcible write back **4**: hybrid	Read-only	Current
hwInfoLunRunningWritePolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.16	Gauge	Operating cache write policy of the LUN. The value can be: **1**: write back **2**: write through **3**: forcible write back **4**: hybrid	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLunPrefetchPolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.17	Gauge	Cache prefetch policy of the LUN. The value can be: **0**: non-prefetch **1**: constant prefetch **2**: variable prefetch **3**: intelligent prefetch	Read-only	Current
hwInfoLunReadCache-Policy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.18	Gauge	Read cache policy of the LUN. The value can be: **1**: permanent **2**: default **3**: reclaim	Read-only	Current
hwInfoLunWriteCache-Policy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.19	Gauge	Write cache policy of the LUN. The value can be: **4**: permanent **5**: default **6**: reclaim	Read-only	Current
hwInfoLunPrefetchValue OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.20	Gauge	Cache prefetch value of the LUN. The cache prefech policy can be: **1**. constant prefetch (unit: KB) **2**. variable prefetch (unit: times) **3**. non-prefetch or intelligent prefetch. If **4294967295** is returned, the value is invalid.	Read-only	Current
hwInfoLunOwnerCon-troller OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.21	OCTET STRING	Owning controller of the LUN	Read-only	Current
hwInfoLunWorkControl-ler OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.22	OCTET STRING	Working controller of the LUN	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLunRelocationPolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.23	Gauge	Data relocation policy of the LUN. The value can be: **0**: no relocation **1**: automatic relocation **2**: highest available **3**: lowest available	Read-only	Current
hwInfoLunIniDistributePolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.24	Gauge	Initial capacity allocation policy of the LUN. The value can be: **0**: automatic **1**: highest performance **2**: performance **3**: capacity	Read-only	Current
hwInfoLunIsAddToLunGroup OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.25	Gauge	Whether the LUN is added to the LUN group. The value can be: **0**: no **1**: yes	Read-only	Current
hwInfoLunDIFSwitch OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.26	Gauge	Whether the DIF function is enabled for the LUN. The value can be: **0**: no **1**: yes	Read-only	Current
hwInfoLunRemoteLUNWWN OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.27	OCTET STRING	WWN of the remote LUN	Read-only	Current
hwInfoLunUsageType OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.28	Gauge	LUN type. The value can be: **0**: internal LUN **1**: external LUN **2**: VVOL LUN **3**: PE LUN	Read-only	Current
hwInfoLunSmartCacheHitRage OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.29	Gauge	Hit ratio of the current intelligent cache partition of the LUN (unit: %)	Read-only	Current

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5.7 Storage Hardware Modules

Note: If the value of an interface parameter is 4294967295 or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.

hwInfoDiskTable

This table is used to check the disk information about a device. The index to this table is **hwInfoDiskID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.1.

Node	Data Type	Meaning	Permission	Status
hwInfoDiskID OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.1	OCTET STRING	Disk ID, table index	Read-only	Current
hwInfoDiskHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.2	Gauge	Disk health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoDiskRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.3	Gauge	Disk running status. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoDiskLocation OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.4	OCTET STRING	Disk location	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskType OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.5	Gauge	Disk type. Possible values are: • 0: FC • 1: SAS • 2: SATA • 3: SSD • 4: NL_SAS • 5: SLC SSD • 6: MLC SSD • 7:FC_SED • 8:SAS_SED • 9:SATA_SED • 10:SSD_SED • 11:NL_SAS_SED • 12:SLC_SSD_SED • 13:MLC_SSD_SED • 14:NVMe_SSD • 15:SMR_SATA • 16:NVMe_SSD_SED • 17:SCM_SSD • 18:SCM_SED	Read-only	Current
hwInfoDiskCapacity OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.6	Counter64	Disk capacity(unit: MB)	Read-only	Current
hwInfoDiskRole OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.7	Gauge	Logical type of a disk. Possible values are: • 1: FREE • 2: MEMBER • 3: SPARE • 4: CACHE	Read-only	Current
hwInfoDiskSpeed OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.8	Gauge	Rotational speed of a disk (unit: rpm)	Read-only	Current
hwInfoDiskInterface-Bandwidth OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.9	Gauge	Disk interface bandwidth (unit: Mbit/s)	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskSectorSize OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.10	Gauge	Sector size of a disk (unit: B)	Read-only	Current
hwInfoDiskTemperature OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.11	Integer	Disk temperature (unit: ºC)	Read-only	Current
hwInfoDiskModel OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.12	OCTET STRING	Disk model	Read-only	Current
hwInfoDiskFirmwareVersion OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.13	OCTET STRING	Firmware version of a disk	Read-only	Current
hwInfoDiskManufacturer OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.14	OCTET STRING	Disk manufacturer	Read-only	Current
hwInfoDiskSerialNumber OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.15	OCTET STRING	Disk SN	Read-only	Current
hwInfoDiskLightStatus OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.16	Gauge	Disk indicator status. Possible values are: • 0: OFF • 1: ON • 2: STEADY ON	Read-only	Current
hwInfoDiskDiskDomainID OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.17	OCTET STRING	ID of a disk domain	Read-only	Current
hwInfoDiskDiskDomainName OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.18	OCTET STRING	Name of a disk domain	Read-only	Current
hwInfoDiskDiskDomainTierID OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.19	OCTET STRING	ID of a disk domain tier	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskCofferDisk OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.20	Gauge	To determine whether a disk is a coffer disk. Possible values are: • 0: NO • 1: YES	Read-only	Current
hwInfoDiskRunTime OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.21	Gauge	Disk operation period (unit: day)	Read-only	Current
hwInfoDiskProgress OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.22	Gauge	Progress of disk reconstruction, copyback, pre-copy, or destruction (unit: %)	Read-only	Current
hwInfoDiskBarCode OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.23	OCTET STRING	Disk barcode	Read-only	Current
hwInfoDiskCapacityUsage OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.24	Gauge	Capacity usage of a disk. (unit: %).	Read-only	Current
hwInfoDiskHealthMark OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.25	Gauge	Health score of a disk. If the value is 255, indicating invalid	Read-only	Current
hwInfoDiskElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.26	OCTET STRING	Electronic label of a disk	Read-only	Current

hwInfoControllerTable

This table is used to check the controller information about a device. The index to this table is **hwInfoControllerID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.2.

Node	Data Type	Meaning	Permission	Status
hwInfoControllerID OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.1	OCTET STRING	Controller ID, table index	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoControllerHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.2	Gauge	Controller health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoControllerRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.3	Gauge	Controller running status. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoControllerCPU OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.4	OCTET STRING	Controller CPU	Read-only	Current
hwInfoControllerLocation OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.5	OCTET STRING	Controller location	Read-only	Current
hwInfoControllerRole OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.6	Gauge	Controller role. Possible values are: • 0: Normal • 1: Master • 2: Slave	Read-only	Current
hwInfoControllerCacheCapacity OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.7	Gauge	Cache capacity of a controller (unit: MB).	Read-only	Current
hwInfoControllerCPUUsage OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.8	Gauge	CPU usage of a controller (unit: %).	Read-only	Current
hwInfoControllerMemoryUsage OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.9	Gauge	Memory usage of the operating system in the controller (unit: %)	Read-only	Current
hwInfoControllerVoltage OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.10	Gauge	Power voltage of a controller (unit: 0.1 V)	Read-only	Current
hwInfoControllerSoftwareVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.11	OCTET STRING	Software version of a controller	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoControllerPCBVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.12	OCTET STRING	PCB version of a controller	Read-only	Current
hwInfoControllerSESVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.13	OCTET STRING	SES version of a controller	Read-only	Current
hwInfoControllerBMCVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.14	OCTET STRING	BMC version of a controller. When the value is empty or "--", indicates unknown or invalid.	Read-only	Current
hwInfoControllerLogicVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.15	OCTET STRING	Logical version of a controller	Read-only	Current
hwInfoControllerBIOSVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.16	OCTET STRING	BIOS version of a controller	Read-only	Current
hwInfoControllerElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.17	OCTET STRING	Electronic label of a controller	Read-only	Current

hwInfoPowerTable

This table is used to check the power supply information about a device. The index to this table is **hwInfoPowerID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.3. When the value is empty or "--", indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwInfoPowerID OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.1	OCTET STRING	ID of a power supply, table index	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoPowerLocation OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.2	OCTET STRING	Location of a power supply	Read-only	Current
hwInfoPowerHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.3	Gauge	Health status of a power supply. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPowerRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.4	Gauge	Operating status of a power supply. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPowerType OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.5	Gauge	Type of a power supply. Possible values are: • 0: DC • 1: AC • 2: HV • 255:Unknown	Read-only	Current
hwInfoPowerManufacturer OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.6	OCTET STRING	Manufacturer of a power supply	Read-only	Current
hwInfoPowerModle OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.7	OCTET STRING	Model of a power supply	Read-only	Current
hwInfoPowerVersion OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.8	OCTET STRING	Version of a power supply	Read-only	Current
hwInfoPowerProduceDate OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.9	OCTET STRING	Production date of a power supply	Read-only	Current
hwInfoPowerSerialNumber OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.10	OCTET STRING	SN of a power supply	Read-only	Current

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hwInfoFanTable

This table is used to check the fan information about a device. The index to this table is **hwInfoFanID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.4. When the value is empty or "--", indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwInfoFanID OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.1	OCTET STRING	Fan ID, table index	Read-only	Current
hwInfoFanLocation OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.2	OCTET STRING	Fan Location	Read-only	Current
hwInfoFanHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.3	Gauge	Health status of a fan. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoFanRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.4	Gauge	Operating status of a fan. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoFanRunningLevel OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.5	Gauge	Operating level of a fan. Possible values are: • 0: Low • 1: Normal • 2: High	Read-only	Current
hwInfoFanElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.6	OCTET STRING	Electronic label of a fan	Read-only	Current

hwInfoBBUTable

This table is used to check the BBU information about a device. The index to this table is **hwInfoBBUID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.5.

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Node	Data Type	Meaning	Permission	Status
hwInfoBBUID OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.1	OCTET STRING	BBU ID, table index	Read-only	Current
hwInfoBBULocation OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.2	OCTET STRING	BBU Location	Read-only	Current
hwInfoBBUHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.3	Gauge	Health status of a BBU. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoBBURunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.4	Gauge	Running status of a BBU. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoBBUType OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.5	Gauge	Type of a BBU. Possible values are: ● 0: BBU	Read-only	Current
hwInfoBBUCurrentVoltage OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.6	Gauge	Current voltage of a BBU (unit: 0.1 V)	Read-only	Current
hwInfoBBUNumberOfDischarges OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.7	Gauge	Discharge times of a BBU	Read-only	Current
hwInfoBBUFirmwareVersion OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.8	OCTET STRING	Firmware version of a BBU	Read-only	Current
hwInfoBBUDeliveredOn OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.9	OCTET STRING	Shipping date of a BBU	Read-only	Current
hwInfoBBUOwningController OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.10	OCTET STRING	Owning controller of a BBU	Read-only	Current
hwInfoBBUElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.11	OCTET STRING	Electronic label of a BBU	Read-only	Current

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hwInfoEnclosureTable

This table is used to check the enclosure information about a device. The index to this table is **hwInfoEnclosureID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.6. When the value is empty or "--", indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwInfoEnclosureID OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.1	OCTET STRING	Enclosure ID, table index	Read-only	Current
hwInfoEnclosureName OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.2	OCTET STRING	Enclosure name	Read-only	Current
hwInfoEnclosureLogicType OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.3	Gauge	Logical type of an enclosure. Possible values are: ● 0: EXP ● 1: CTRL ● 2: DSW ● 3: MSW ● 4: SVP	Read-only	Current
hwInfoEnclosureHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.4	Gauge	Health status of an enclosure. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoEnclosureRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.5	Gauge	Operating status of an enclosure. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoEnclosureLocation OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.6	OCTET STRING	Enclosure location	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoEnclosureType OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.7	Gauge	Enclosure type. Possible values are: 0:BMC Controller 1:2 U 2-controller 12-slot 3.5-inch 6 Gbit/s SAS controller enclosure 2:2 U 2-controller 24-slot 2.5-inch 6 Gbit/s SAS controller enclosure 16:2 U 12-slot 3.5-inch 6 Gbit/s SAS disk enclosure 17:2 U 24-slot 2.5-inch 6 Gbit/s SAS disk enclosure 18:4 U 24-slot 3.5-inch 6 Gbit/s SAS disk enclosure 19:4 U 24 Slot 3.5 FC Disks Enclosure 20:1 U PCIE Switch Enclosure 21:4 U 75-slot 3.5-inch 6 Gbit/s SAS disk enclosure 22:Service Processor Enclosure 23:2 U 2-controller 12-slot 3.5-inch 6 Gbit/s SAS controller enclosure 24:2 U 25-slot 2.5-inch 6 Gbit/s SAS disk enclosure 25:4 U 24-slot 3.5-inch 6 Gbit/s SAS disk enclosure 26:2 U 2-controller 25-slot 2.5-inch 6 Gbit/s SAS controller enclosure 37:2 U 2-controller 12-slot 3.5-inch 6 Gbit/s SAS controller enclosure 38:2 U 2-controller 25-slot 2.5-inch 6 Gbit/s SAS controller enclosure 39:4 U 75-slot 3.5-inch 12 Gbit/s SAS disk enclosure 40:2 U 2-controller 25-slot 2.5-inch 12 Gbit/s SAS controller enclosure	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
		65:2 U 25-slot 2.5-inch 12 Gbit/s SAS disk enclosure 66:4 U 24-slot 3.5-inch 12 Gbit/s SAS disk enclosure 67:2 U 25-slot 2.5-inch SAS disk enclosure 69:4 U 24-slot 3.5-inch SAS disk enclosure 96:3 U 2-controller controller enclosure 97:6 U 4-controller controller enclosure 98:2 U 25 Slot 2.5 SSD Disks Enclosure 99:2 U 25-slot 2.5-inch active NVMe controller enclosure 101:2 U 25-slot 2.5-inch SSD NVMe disk enclosure 112:4 U 4-controller controller enclosure 113:2 U 2-controller 25-slot 2.5-inch SAS controller enclosure 114:2 U 2-controller 12-slot 3.5-inch SAS controller enclosure 115:2 U 2-controller 36-slot NVMe controller enclosure 116:2 U 2-controller 25-slot 2.5-inch SAS controller enclosure 117:2 U 2-controller 12-slot 3.5-inch SAS controller enclosure 118:2 U 25-slot 2.5-inch smart SAS disk enclosure 119:2 U 12-slot 3.5-inch smart SAS disk enclosure 120:2 U 36-slot smart NVMe disk enclosure		

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Node	Data Type	Meaning	Permission	Status
		122:2U 2-controller 25-slot 2.5-inch NVMe controller enclosure 130:2U 2-controller 25-slot 2.5-inch SAS controller enclosure 131:2U 2-controller 12-slot 3.5-inch SAS controller enclosure		
hwInfoEnclosureTemperature OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.8	Integer	Enclosure temperature (unit: ºC)	Read-only	Current
hwInfoEnclosureSN OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.9	OCTET STRING	Enclosure SN	Read-only	Current
hwInfoEnclosureMAC OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.10	OCTET STRING	MAC address of an enclosure	Read-only	Current
hwInfoEnclosureHeight OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.11	Gauge	Enclosure height (unit: U)	Read-only	Current
hwInfoEnclosureExpansionDepth OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.12	Gauge	Number of cascaded enclosures	Read-only	Current
hwInfoEnclosureElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.13	OCTET STRING	Electronic label of an enclosure	Read-only	Current

hwInfoPortComTable

This table is used to check the serial port information about a device. The index to this table is **hwInfoPortComID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.7.

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Node	Data Type	Description	Permission	Status
hwInfoPortComID OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.1	OCTET STRING	ID of a serial port, table index	Read-only	Current
hwInfoPortComLocation OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.2	OCTET STRING	Location of a serial port	Read-only	Current
hwInfoPortComHealth-Status OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.3	Gauge	Health status of a serial port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortComRunning-Status OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.4	Gauge	Operating status of a serial port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortComType OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.5	Gauge	Logical type of a serial port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current

hwInfoPortEthTable

This table is used to check the Ethernet port information about a device. The index to this table is **hwInfoPortEthID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.8. When the value is empty or "--", "4294967295" or "F", indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortEthID OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.1	OCTET STRING	ID of an Ethernet port, table index	Read-only	Current
hwInfoPortEthLocation OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.2	OCTET STRING	Location of an Ethernet port	Read-only	Current
hwInfoPortEthHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.3	Gauge	Health status of an Ethernet port. For details, see **HEALTH_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortEthRunning-Status OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.4	Gauge	Operating status of an Ethernet port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortEthType OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.5	Gauge	Logical type of an Ethernet port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current
hwInfoPortEthIPv4Address OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.6	OCTET STRING	IPv4 address of an Ethernet port	Read-only	Current
hwInfoPortEthSubnet-Mask OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.7	OCTET STRING	IPv4 mask of an Ethernet port	Read-only	Current
hwInfoPortEthIPv4Gateway OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.8	OCTET STRING	IPv4 gateway of an Ethernet port	Read-only	Current
hwInfoPortEthIPv6Address OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.9	OCTET STRING	IPv6 address of an Ethernet port	Read-only	Current
hwInfoPortEthIPv6PrefixLength OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.10	OCTET STRING	IPv6 mask of an Ethernet port	Read-only	Current
hwInfoPortEthIPv6Gateway OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.11	OCTET STRING	IPv6 gateway of an Ethernet port	Read-only	Current
hwInfoPortEthMAC OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.12	OCTET STRING	MAC address of an Ethernet port	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortEthRole OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.13	Gauge	Role of an Ethernet port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current
hwInfoPortEthMode OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.14	Gauge	Auto-negotiation mode of an Ethernet port. Possible values are: 1: HALF_DUPLEX 2: DUPLEX 3: AUTO If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortEthMTU OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.15	Gauge	Maximum transmission unit (MTU) of an Ethernet port (unit: B). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortEthWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.16	Gauge	Working rate of an Ethernet port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortEthBondName OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.17	OCTET STRING	Bond name of an Ethernet port	Read-only	Current
hwInfoPortEthiSCSIPort OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.18	OCTET STRING	iSCSI port number of an Ethernet port. The port is occupied when iSCSI is enabled. The default port number is 3260.	Read-only	Current
hwInfoPortEthiSCSIName OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.19	OCTET STRING	iSCSI name of an Ethernet port	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortEthErrorPackets OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.20	Gauge	Number of error packets of an Ethernet port	Read-only	Current
hwInfoPortEthLostPackages OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.21	Gauge	Number of lost packets of an Ethernet port	Read-only	Current
hwInfoPortEthStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.22	OCTET STRING	Time when the bit error statistics of an Ethernet port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

hwInfoPortFCTable

This table is used to check the Fibre Channel port information about a device. The index to this table is **hwInfoPortFCID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.9.

Node	Data Type	Description	Permission	Status
hwInfoPortFCID OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.1	OCTET STRING	ID of a Fibre Channel port, table index	Read-only	Current
hwInfoPortFCLocation OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.2	OCTET STRING	Location of a Fibre Channel port	Read-only	Current
hwInfoPortFCHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.3	Gauge	Health status of a Fibre Channel port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortFCRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.4	Gauge	Operating status of a Fibre Channel port. For details, see **RUNNING_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortFCType OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.5	Gauge	Logical type of a Fibre Channel port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current
hwInfoPortFCWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.6	Gauge	Working rate of a Fibre Channel port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortFCConfiguredSpeed OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.7	Gauge	Configured rate of a Fibre Channel port (unit: Mbit/s). **0** indicates auto-negotiation. **4294967295** indicates invalid.	Read-only	Current
hwInfoPortFCWWN OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.8	OCTET STRING	World wide name (WWN) of a Fibre Channel port	Read-only	Current
hwInfoPortFCRole OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.9	Gauge	Role of a Fibre Channel port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current
hwInfoPortFCSFPStatus OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.10	Gauge	Status of a Fibre Channel port. Possible values are: 0: Not Exist 1: Offline 2: Online 4294967295: Invalid	Read-only	Current
hwInfoPortFCWorkingMode OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.11	Gauge	Operating mode of a Fibre Channel port. Possible values are: 0: Fabric 1: FC-AL 2: Point To Point 3: Auto 4294967295: Invalid	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortFCConfigur-edMode OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.12	Gauge	Configured mode of a Fibre Channel port. Possible values are: 0: Fabric 1: FC-AL 2: P2P 3: Auto 4294967295: Invalid	Read-only	Current
hwInfoPortFCFloginDe-layTimes OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.13	Gauge	Flogin delay of a Fibre Channel port (unit: ms)	Read-only	Current
hwInfoPortFCLostSignals OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.14	Gauge	Number of Loss of Signal errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCLinkEr-rorsCodes OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.15	Gauge	Number of Bad RX Char errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCLostSynch-ronizations OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.16	Gauge	Number of Loss of Sync errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCFailedCon-nections OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.17	Gauge	Number of Link Fail errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.18	OCTET STRING	Time when the bit error statistics of a Fibre Channel port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

hwInfoPortFCoETable

This table is used to check the FCoE port information about a device. The index to this table is **hwInfoPortFCoEID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.10.

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Node	Data Type	Description	Permission	Status
hwInfoPortFCoEID OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.1	OCTET STRING	ID of an FCoE port, table index	Read-only	Current
hwInfoPortFCoELocation OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.2	OCTET STRING	Location of an FCoE port	Read-only	Current
hwInfoPortFCoEHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.3	Gauge	Health status of an FCoE port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortFCoERunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.4	Gauge	Operating status of an FCoE port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortFCoEType OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.5	Gauge	Logical type of an FCoE port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current
hwInfoPortFCoEWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.6	Gauge	Working rate of an FCoE port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortFCoEWWN OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.7	Gauge	WWN of an FCoE port	Read-only	Current
hwInfoPortFCoERole OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.8	OCTET STRING	Role of an FCoE port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current
hwInfoPortFCoESFPStatus OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.9	Gauge	Status of the optical module in an FCoE port. Possible values are: 0: NO_EXIST 1: OFFLINE 2: ONLINE 4294967295: Invalid	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortFCoEErrorPackets OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.10	Gauge	Number of error packets of an FCoE port	Read-only	Current
hwInfoPortFCoELostPackets OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.11	Gauge	Number of lost packets of an FCoE port	Read-only	Current
hwInfoPortFCoEOverFlowedPackets OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.12	Gauge	Number of overflowed packets of an FCoE port	Read-only	Current
hwInfoPortFCoEStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.10.1.13	OCTET STRING	Time when the bit error statistics of an FCoE port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

hwInfoPortPCIETable

This table is used to check the PCIe port information about a device. The index to this table is **hwInfoPortPCIEID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.11. The value of performance statistics entry is "4294967295" or "F", indicates an unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortPCIEID OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.1	OCTET STRING	ID of a PCIe port, table index	Read-only	Current
hwInfoPortPCIELocation OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.2	OCTET STRING	Location of a PCIe port	Read-only	Current
hwInfoPortPCIEHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.3	Gauge	Health status of a PCIe port. For details, see **HEALTH_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortPCIERunning-Status OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.4	Gauge	Operating status of a PCIe port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortPCIESpeed OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.5	Gauge	Data transmission rate of a PCIe port (unit: Mbit/s)	Read-only	Current
hwInfoPortPCIECurrent-PeerPortID OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.6	OCTET STRING	ID of the port connecting to a PCIe port	Read-only	Current
hwInfoPortPCIESuggest-PeerPortID OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.7	OCTET STRING	ID of the port that is supposed to connect to a PCIe port	Read-only	Current
hwInfoPortPCIELostSignals OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.8	Gauge	Number of bit errors of a PCIe port	Read-only	Current
hwInfoPortPCIEECRCError OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.9	Gauge	Number of ECRC errors of a PCIe port	Read-only	Current
hwInfoPortPCIEBadTlp OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.10	Gauge	Number of bad TLPs of a PCIe port	Read-only	Current
hwInfoPortPCIEBadDllp OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.11	Gauge	Number of bad DLLPs of a PCIe port	Read-only	Current
hwInfoPortPCIERecvError OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.12	Gauge	Number of receiver errors of a PCIe port	Read-only	Current
hwInfoPortPCIEChipEccError OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.13	Gauge	Number of chip ECC errors of a PCIe port	Read-only	Current
hwInfoPortPCIEStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.14	OCTET STRING	Time when the bit error statistics of a PCIe port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

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hwInfoPortSASTable

This table is used to check the SAS port information about a device. The index to this table is **hwInfoPortSASID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.12. The value of performance statistics entry is "4294967295" or "F", indicates an unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortSASID OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.1	OCTET STRING	ID of a SAS port, table index	Read-only	Current
hwInfoPortSASLocation OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.2	OCTET STRING	Location of a SAS port	Read-only	Current
hwInfoPortSASHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.3	Gauge	Health status of a SAS port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortSASRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.4	Gauge	Operating status of a SAS port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortSASType OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.5	Gauge	Logical type of a SAS port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current
hwInfoPortSASWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.6	Gauge	Working rate of a SAS port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortSASWWN OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.7	OCTET STRING	WWN of a SAS port	Read-only	Current
hwInfoPortSASRole OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.8	Gauge	Role of a SAS port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortSASInvalidDword OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.9	Gauge	Number of invalid DWORDs of a SAS port	Read-only	Current
hwInfoPortSASConsistErrors OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.10	Gauge	Number of consistency errors of a SAS port	Read-only	Current
hwInfoPortSASLossOfDword OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.11	Gauge	Number of lost DWORDs of a SAS port	Read-only	Current
hwInfoPortSASPHYResetErrors OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.12	Gauge	Number of physical reset errors of a SAS port	Read-only	Current
hwInfoPortSASStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.13	OCTET STRING	Time when the bit error statistics of a SAS port starts. If "--" is returned, this parameter is invalid.	Read-only	Current
hwInfoPortSASEnabled OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.14	Gauge	To determine whether a SAS port is enabled. Possible values are: 0: No 1: Yes	Read-only	Current

hwInfoInterfaceModuleTable

This table is used to check the interface module information about a device. The index to this table is **hwInfoInterfaceModuleID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.13.

Node	Data Type	Description	Permission	Status
hwInfoInterfaceModuleID OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.1	OCTET STRING	ID of an interface module, table index	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoInterfaceModuleLocation OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.2	OCTET STRING	Location of an interface module	Read-only	Current
hwInfoInterfaceModuleHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.3	Gauge	Health status of an interface module. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoInterfaceModuleRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.4	Gauge	Operating status of an interface module. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoInterfaceModuleModel OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.5	Gauge	Type of an interface module. For details, see **INTF_MODEL_E**.	Read-only	Current
hwInfoInterfaceModuleLogicVersion OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.6	OCTET STRING	Logical version of an interface module, When the value is "--", indicates unknown.	Read-only	Current
hwInfoInterfaceModulePCBVersion OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.7	OCTET STRING	PCB version of an interface module	Read-only	Current
hwInfoInterfaceModuleElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.8	OCTET STRING	Electronic label of an interface module	Read-only	Current

hwInfoExpBoardTable

This table is used to check the expansion module information about a device. The index to this table is **hwInfoExpBoardID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.14.

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Node	Data Type	Description	Permission	Status
hwInfoExpBoardID OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.1	OCTET STRING	ID of an expansion module, table index	Read-only	Current
hwInfoExpBoardLocation OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.2	OCTET STRING	Location of an expansion module	Read-only	Current
hwInfoExpBoardHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.3	Gauge	Health status of an expansion module. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoExpBoardRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.4	Gauge	Operating status of an expansion module. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoExpBoardModel OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.5	Gauge	Model of an expansion module. Possible values are: ● 0: SAS ● 1: FC	Read-only	Current
hwInfoExpBoardLogicVersion OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.6	OCTET STRING	Logical version of an expansion module, When the value is "--", indicates unknown	Read-only	Current
hwInfoExpBoardPCBVersion OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.7	OCTET STRING	PCB version of an expansion module	Read-only	Current
hwInfoExpBoardSESVersion OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.8	OCTET STRING	SES version of an expansion module	Read-only	Current
hwInfoExpBoardELabel OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.9	OCTET STRING	Electronic label of an expansion module	Read-only	Current

hwInfoCPUTable

This table is used to check the CPU information about a device. The index to this table is **hwInfoCPUID**.

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The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.15.

Node	Data Type	Description	Permission	Status
hwInfoCPUID OID:1.3.6.1.4.1.34774.4.1.23.5.15.1.1	OCTET STRING	CPU ID, table index	Read-only	Current
hwInfoCPUCoreTemp OID:1.3.6.1.4.1.34774.4.1.23.5.15.1.2	Unsigned32	CPU temperature (unit: ℃)	Read-only	Current

5.8 NAS Information Query Module

hwInfoShareNFSTable

This table is used to check the NFS share information about a device. The index to this table is **hwInfoShareNFSShareID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.1.

Node	Data Type	Description	Permission	Status
hwInfoShareNFSShareID OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.1	OCTET STRING	Share ID, table index	Read-only	Current
hwInfoShareNFSFileSystemID OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.2	OCTET STRING	File system ID	Read-only	Current
hwInfoShareNFSDescription OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.3	OCTET STRING	Description	Read-only	Current
hwInfoShareNFSLocalPath OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.4	OCTET STRING	Share path	Read-only	Current

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hwInfoShareCIFSTable

This table is used to check the CIFS share information about a device. The index to this table is **hwInfoShareCIFSID**.

The the OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.2.

Node	Data Type	Description	Permission	Status
hwInfoShareCIFSID OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.1	OCTET STRING	Share ID, table index	Read-only	Current
hwInfoShareCIFSName OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.2	OCTET STRING	Object name	Read-only	Current
hwInfoShareCIFSFileSystemID OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.3	OCTET STRING	File system ID	Read-only	Current
hwInfoShareCIFSDescription OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.4	OCTET STRING	Description	Read-only	Current
hwInfoShareCIFSLocalPath OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.5	OCTET STRING	Share path	Read-only	Current
hwInfoShareCIFSOplockEnabled OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.6	Gauge	Optimization lock switch • 0: No • 1: Yes	Read-only	Current
hwInfoShareCIFSNotifyEnabled OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.7	Gauge	Notification switch • 0: No • 1: Yes	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoShareCIFSContAvailableEna OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.8	Gauge	Whether failover is enabled • 0: No • 1: Yes	Read-only	Current

hwInfoSharePermsNFSTable

This table is used to check the NFS share permission of a device. The index to this table is **hwInfoSharePermsNFSID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.3.

Node	Data Type	Description	Permission	Status
hwInfoSharePermsNFSID OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoSharePermsNFSAccessName OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.2	OCTET STRING	Access name	Read-only	Current
hwInfoSharePermsNFSShareID OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.3	OCTET STRING	Share ID	Read-only	Current
hwInfoSharePermsNFSAccessType OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.4	Gauge	Access type • 0: read-only • 1: read and write • 4: read and write but cannot be deleted or renamed, supported by massive storage only)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoSharePermsNFS-SyncEnabled OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.5	Gauge	Synchronization switch • 0: Data is synchronized • 1: Data is not synchronized	Read-only	Current
hwInfoSharePermsNF-SAllSquashEna OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.6	Gauge	Whether the GID and UID of the shared directory are mapped as anonymous users • 0: yes • 1: no	Read-only	Current
hwInfoSharePermsNFS-RootSquashEna OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.7	Gauge	Permission squash of user root • 0: rootSquash • 1: no rootSquash	Read-only	Current

hwInfoSharePermsCIFSTable

This table is used to check the CIFS share permission of a device. The index to this table is **hwInfoSharePermsCIFSID**.

The prefix to the OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.4. Obtain the actual OID of each node in the MIB file.

Node	Data Type	Description	Permission	Status
hwInfoSharePermsCIFSID OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoSharePermsCIFSAc-cessName OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.2	OCTET STRING	Name of a user or user group	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoSharePermsCIFS-ShareID OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.3	OCTET STRING	Share ID	Read-only	Current
hwInfoSharePermsCIFS-DomainType OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.4	Gauge	Domain type • 0: AD • 1: LDAP • 2: local user	Read-only	Current
hwInfoSharePermsCIF-SPermsType OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.5	Gauge	Permission type • 0: read • 1: full control • 2: forbidden • 3: default • 4: read and write (cannot be deleted, renamed) • 5: read and write, being used in unified storage but not massive storage	Read-only	Current

hwInfoLogicalPortTable

This table is used to check the logical port information about a device. The index to this table is **hwInfoLogicalPortID**.

The the OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.5.

Node	Data Type	Description	Permission	Status
hwInfoLogicalPortID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoLogicalPorttName OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.2	OCTET STRING	Object name	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortRunStatus OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.3	Gauge	Operating status (for details, see **RUNNING_STATUS_E**)	Read-only	Current
hwInfoLogicalPortIPv4Addr OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.4	OCTET STRING	IPv4 address	Read-only	Current
hwInfoLogicalPortIPv4Mask OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.5	OCTET STRING	IPv4 mask	Read-only	Current
hwInfoLogicalPortIPv4Gateway OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.6	OCTET STRING	IPv4 gateway	Read-only	Current
hwInfoLogicalPortIPv6Addr OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.7	OCTET STRING	IPv6 address	Read-only	Current
hwInfoLogicalPortIPv6Mask OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.8	OCTET STRING	IPv6 mask	Read-only	Current
hwInfoLogicalPortIPv6Gateway OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.9	OCTET STRING	IPv6 gateway	Read-only	Current
hwInfoLogicalPortRole OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.10	Gauge	Role of the logical port • 0: unknown • 1: management • 2: service • 3: all	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortSupportProt OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.11	Gauge	Protocol: • 0: none • 1: NFS • 2: CIFS • 3: NFS + CIFS • 4: iSCSI • 8: FC/FCoE • 16: HTTPS • 32: NDMP • 35: NFS+CIFS+NDMP • 51: HTTPS+NFS+CIFS+NDMP • 64: NVMe	Read-only	Current
hwInfoLogicalPortHomePortType OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.12	Gauge	Type of the parent port: • 1: ETH • 2: FC • 3: FCoE • 4: SAS • 5: PCIe • 6: IB • 7: bond • 8: VLAN • 9: LIF	Read-only	Current
hwInfoLogicalPortHomePortID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.13	OCTET STRING	ID of the parent port	Read-only	Current
hwInfoLogicalPortOwnerCtrlID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.14	OCTET STRING	ID of the parent node	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortCurrPortType OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.15	Gauge	Port type • 1: ETH • 2: FC • 3: FCoE • 4: SAS • 5: PCIe • 6: IB • 7: bond • 8: VLAN • 9: LIF	Read-only	Current
hwInfoLogicalPortCurrPortID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.16	OCTET STRING	Port ID	Read-only	Current
hwInfoLogicalPortWorkCtrlID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.17	OCTET STRING	Node ID	Read-only	Current
hwInfoLogicalPortActState OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.18	Gauge	Active or not • 0: No • 1: Yes	Read-only	Current
hwInfoLogicalPortAddrFamily OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.19	Gauge	Address type • 0: IPv4 • 1: IPv6	Read-only	Current
hwInfoLogicalPortIsPrivate OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.20	Gauge	Configurable or not • 0: No • 1: Yes	Read-only	Current
hwInfoLogicalPortFailOVGID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.21	OCTET STRING	ID of the owning failover group	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortFailOVEnable OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.22	Gauge	Whether failover is enabled • 0: No • 1: Yes	Read-only	Current
hwInfoLogicalPortFailBackMode OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.23	Gauge	Failover mode: • 1: manual • 2: automatic	Read-only	Current
hwInfoLogicalPortFailOVGName OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.24	OCTET STRING	Name of the owning failover group	Read-only	Current

hwInfoFileSysTable

This table is used to check the file system information about a device. The index to this table is **hwInfoFileSysID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.6.

Node	Data Type	Description	Permission	Status
hwInfoFileSysID OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoFileSysName OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.2	OCTET STRING	File system name	Read-only	Current
hwInfoFileSysHeathStatus OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.3	Gauge	Health status (for details, see **HEALTH_STATUS_E**)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoFileSysRunningStatus OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.4	Gauge	Operating status (for details, see **RUNNING_STATUS_E**)	Read-only	Current
hwInfoFileSysDescription OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.5	OCTET STRING	Description	Read-only	Current
hwInfoFileSysSubType OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.6	Gauge	Subtype: • 0: Normal • 1: WORM	Read-only	Current
hwInfoFileSysAllocType OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.7	Gauge	Space allocation type: • 0: thick • 1: thin	Read-only	Current
hwInfoFileSysCapacity OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.8	Counter64	Configured capacity (unit: MB)	Read-only	Current
hwInfoFileSysSnapshotReservePer OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.9	Gauge	Space reserved for snapshot (unit: %)	Read-only	Current
hwInfoFileSysSnapshotUseCapacity OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.10	Counter64	Capacity used by snapshots (unit: MB)	Read-only	Current
hwInfoFileSysSectorSize OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.11	Gauge	Block size (unit: byte)	Read-only	Current
hwInfoFileSysOwningContrller OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.12	OCTET STRING	Owning controller	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoFileSysWorking-Controller OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.13	OCTET STRING	Working controller	Read-only	Current
hwInfoFileSysIOPriotiry OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.14	Gauge	File system I/O priority: • 1: low • 2: medium • 3: high	Read-only	Current
hwInfoFileSysEnableCompression OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.15	Gauge	Whether compression is enabled • 0: No • 1: Yes	Read-only	Current
hwInfoFileSysCompression OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.16	Gauge	Compression algorithm: • 0: fast • 1: slow	Read-only	Current
hwInfoFileSysIsShowSnapDir OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.17	Gauge	Snapshot directory visible or not • 0: No • 1: Yes	Read-only	Current
hwInfoFileSysAvailableCapacity OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.18	Counter64	Available file system capacity (unit: MB)	Read-only	Current
hwInfoFileSysAvAndAllcCapRatio OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.19	Gauge	Percentage of available file system used capacity to initial capacity	Read-only	Current
hwInfoFileSysSCCachedSize OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.20	Counter64	SmartCache size (unit: KB)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoFileSysSCHitRage OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.21	Gauge	SmartCache hit ratio (unit: %)	Read-only	Current
hwInfoFileSysCompressSavedRatio OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.22	Gauge	Compression ratio (unit: %)	Read-only	Current

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6 Appendix

6.1 Common Enumerations

6.1 Common Enumerations

6.1.1 Definition of Enum Values (HEALTH_STATUS_E)

Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	--	7	Bit Errors Found	14	Invalid
1	Normal	8	Consistent	15	Write Protect
2	Fault	9	Inconsistent	16	single link
3	Pre-Fail	10	Busy	17	No Redundant Link
4	Partially Broken	11	No Input	18	Offline
5	Degraded	12	Low Battery		
6	Bad Sectors Found	13	Single Link Fault		

6.1.2 Definition of Enum Values (RUNNING_STATUS_E)

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	--	48	Charging	96	Partition Migrating
1	Normal	49	Charging completed	97	Mount
2	Running	50	Discharging	98	Umount
3	Not running	51	Upgrading	99	INSTALLING
4	Not existed	52	Power Lost	100	To Be Synchronized
5	Sleep in high temperature	53	Initializing	101	Connecting
6	Starting	54	Apply change	102	Service Switching
7	Power failure rotection	55	Online disable	103	Power-on failed
8	Spin down	56	Offline disable	104	REPAIRING
9	Started	57	Online frozen	105	abnormal
10	Link Up	58	Offline frozen	106	Deleting
11	Link Down	59	Closed	107	Modifying
12	Powering on	60	Removing	108	Running(clearing data)
13	Powered off	61	In service	109	Running(synchronizing data)
14	Pre-copy	62	Out of service	110	Standby
15	Copyback	63	Running normal	111	STOPPING
16	Reconstruction	64	Running fail	112	RESTORE FAULT
17	Expansion	65	Running success	113	Cut Over
18	Unformatted	66	Running success	114	Erasing
19	Formatting	67	Running failed	115	Verifying
20	Unmapped	68	Waiting	119	Creating
21	Initial synchronizing	69	Canceling		
22	Consistent	70	Canceled		
23	Synchronizing	71	About to synchronize		

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
24	Synchronized	72	Synchronizing data		
25	Unsynchronized	73	Failed to synchronize		
26	Split	74	Fault		
27	Online	75	Migrating		
28	Offline	76	Migrated		
29	Locked	77	Activating		
30	Enabled	78	Deactivating		
31	Disabled	79	Start failed		
32	Balancing	80	Stop failed		
33	To be recovered	81	Decommissioning		
34	Interrupted	82	Decommissioned		
35	Invalid	83	Recommissioning		
36	Not start	84	Replacing node		
37	Queuing	85	Scheduling		
38	Stopped	86	Pausing		
39	Copying	87	Suspending		
40	Completed	88	Suspended		
41	Paused	89	Overload		
42	Reverse synchronizing	90	To be switch		
43	Activated	91	Switching		
44	Restore	92	To be cleanup		
45	Inactive	93	Forced start		
46	Idle	94	Error		
47	Powering off	95	Job completed		

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6.1.3 Definition of Enum Values (PORT_LOGIC_TYPE_E)

Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	HOST	6	MAINTENANCE_SRV		
1	EXP	7	BACKUP_MGR		
2	MNGT	8	PRODUCT_STORAGE		
3	INNER	9	BACKUP_STORAGE		
4	MAINTENANCE	10	ETH_NOT_CONFIG		
5	MNGT_SRV	11	IP_SCALE_OUT		

6.1.4 Definition of Enum Values (INTF_MODEL_E)

Enum Values	English Term	Enum Values	English Term
1	4 x 4 G FC Interface Module	2316	2 ports FE 100 Gbit/s ETH I/O module
2	2 x 4 G FC Interface Module	2317	2 ports BE 100 Gbit/s RDMA I/O module
3	2 x GE Electrical Interface Module	2318	2 ports SO 100 Gbit/s RDMA I/O module
4	4 x SAS I Interface Module	2319	2 ports FE 40 Gbit/s ETH I/O module
5	Swapped FC Interface Module	2320	2 ports FE 100 Gbit/s ETH I/O module
6	2 x 10GE Optical Interface Module	2321	2 ports BE 100 Gbit/s RDMA I/O module
7	1 x 8 G FC Optical Interface Module	2322	2 ports SO 100 Gbit/s RDMA I/O module
8	2 x 8 G FC Optical Interface Module	2323	4 ports FE 10 Gbit/s ROCE I/O module

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Enum Values	English Term	Enum Values	English Term
9	2 x SAS II Electrical Interface Module	2324	4 ports FE 25 Gbit/s ROCE I/O module
10	4 x SAS II Electrical Interface Module	2325	4 ports FE 10 Gbit/s ROCE I/O module
11	2 x FC + 2 x GE Interface Module	2326	4 ports FE 25 Gbit/s ROCE I/O module
12	4 x GE Electrical Interface Module	2327	2 ports FE 40 Gbit/s ROCE I/O module
13	4 x 8 G FC Optical Interface Module	2328	2 ports FE 100 Gbit/s ROCE I/O module
16	4 x mini-SAS Interface Module	2329	2 ports FE 40 Gbit/s ROCE I/O module
21	4 x 10 G FCoE Optical Interface Module	2330	2 ports FE 100 Gbit/s ROCE I/O module
22	2 x 6 G SAS Interface Module	2331	4 ports FE 10 Gbit/s ETH I/O module
23	2 x 6 G SAS Interface Module	2332	4 ports FE 10 Gbit/s ETH I/O module
24	Management Board	2333	4 ports FE 8 Gbit/s Fibre Channel I/O module
25	4 x 10GE Interface Module	2334	4 ports FE 16 Gbit/s Fibre Channel I/O module
26	PCIe Interface Module	2335	4 ports FE 32 Gbit/s Fibre Channel I/O module
27	2x10G Optical Interface Module	2336	4 ports FE 10 Gbit/s ETH I/O module
28	4xGE Optical Interface Module	2337	4 ports FE 25 Gbit/s ETH I/O module
29	2 x 16 G FC Optical Interface Module	2338	4 ports SO 25 Gbit/s RDMA I/O module
30	4 x 12SAS QSFP Interface Module	2339	4 ports FE 10 Gbit/s ROCE I/O module
31	4 x 10GE Electrical Interface Module	2340	4 ports FE 25 Gbit/s ROCE I/O module
32	2 x 40GE Infiniband Interface Module	2341	4 ports FE 8 Gbit/s Fibre Channel I/O module

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Enum Values	English Term	Enum Values	English Term
33	2 port 4*14Gb IB I/O module	2342	4 ports FE 16 Gbit/s Fibre Channel I/O module
34	4 x GE Electrical Interface Module	2343	4 ports FE 32 Gbit/s Fibre Channel I/O module
35	Smart ACC module	2344	4 ports FE 10 Gbit/s ETH I/O module
36	4x10GE Electrical Interface Module	2345	4 ports FE 25 Gbit/s ETH I/O module
37	4 port SmartIO I/O module	2346	4 ports FE 10 Gbit/s ROCE I/O module
38	8x8G FC Optical Interface Module	2347	4 ports FE 25 Gbit/s ROCE I/O module
39	4x8G FC Optical Interface Module	2348	2 ports FE 40 Gbit/s ETH I/O module
40	4x16G FC Optical Interface Module	2349	2 ports FE 100 Gbit/s ETH I/O module
41	12 port 4*12Gb SAS Back-End Interconnect I/O Module	2350	2 ports BE 100 Gbit/s RDMA I/O module
42	12 port SmartIO I/O module	2351	2 ports SO 100 Gbit/s RDMA I/O module
43	RoCE 10G/25G Interface Module	2352	2 ports FE 40 Gbit/s ROCE I/O module
44	2-port PCIe 3.0 Interface Module	2353	2 ports FE 100 Gbit/s ROCE I/O module
55	2-port 8 Gbit/s PCIe scale-up interface module	2354	2 ports FE 40 Gbit/s ETH I/O module
58	8x16G FC Optical Interface Module	2355	2 ports FE 100 Gbit/s ETH I/O module
65	4 ports SmartIO I/O module	2356	2 ports BE 100 Gbit/s RDMA I/O module
66	4 ports SmartIO I/O module	2357	2 ports SO 100 Gbit/s RDMA I/O module
67	2 ports 100Gb/40Gb ETH I/O module	2358	2 ports FE 40 Gbit/s ROCE I/O module
68	2 ports 100Gb/40Gb ETH I/O module	2359	2 ports FE 100 Gbit/s ROCE I/O module

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Enum Values	English Term	Enum Values	English Term
71	4 ports 10Gb ETH I/O module	2360	4 ports FE 10 Gbit/s ETH I/O module
516	4 ports FE 1 Gbit/s ETH I/O module	2361	4 ports SO 25 Gbit/s RDMA I/O module
518	4 ports BE 12 Gbit/s SAS I/O module	2362	2 ports SO 100 Gbit/s RDMA I/O module
529	AI Accelerator Card	2363	2 ports SO 100 Gbit/s RDMA I/O module
535	AI Accelerator Card	4133	System Management Module
537	4 ports FE 1 Gbit/s ETH I/O module	4134	System Management Module
538	4 ports BE 12 Gbit/s SAS I/O module	4294967295	unknown
580	4 ports FE 1 Gbit/s ETH I/O module		
583	4 ports BE 12 Gbit/s SAS V2 I/O module		
601	4 ports FE 1 Gbit/s ETH I/O module		
2304	4 ports FE 8 Gbit/s Fibre Channel I/O module		
2305	4 ports FE 16 Gbit/s Fibre Channel I/O module		
2306	4 ports FE 32 Gbit/s Fibre Channel I/O module		
2307	4 ports FE 10 Gbit/s ETH I/O module		
2308	4 ports FE 25 Gbit/s ETH I/O module		
2309	4 ports SO 25 Gbit/s RDMA I/O module		
2310	4 ports FE 8 Gbit/s Fibre Channel I/O module		
2311	4 ports FE 16 Gbit/s Fibre Channel I/O module		

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Enum Values	English Term	Enum Values	English Term
2312	4 ports FE 32 Gbit/s Fibre Channel I/O module		
2313	4 ports FE 10 Gbit/s ETH I/O module		
2314	4 ports FE 25 Gbit/s ETH I/O module		
2315	2 ports FE 40 Gbit/s ETH I/O module		

6.1.5 Definition of Enum Values (PRODUCT_MODE_E)

Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	V1500	63	5600 V3	131	18510F V5
1	V1800	64	5800 V3	132	18810 V5
2	S2100	65	T3200 G3	133	18810F V5
3	S2300	66	WUSHAN_SQL	134	5210 V5 Enhanced
4	S5100	67	WUSHAN_HD	135	5210F V5 Enhanced
5	S5300	68	5500 V3	139	5110 V5 Enhanced
6	S5500	69	2600 V3	140	5110F V5 Enhanced
7	S5600	70	5300 V3	141	2600 V5
8	VIS	71	2800 V3	143	2210
9	S2600	72	18500 V3	144	5120 V5
10	S6800E	73	18800 V3	145	5220 V5
11	V1500N	74	High-Density Controller Enclosure	147	2220
12	S2300E	78	2200 V3	401	N5000
13	COMMON	79	5100 V3	800	VIS8200

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
15	N8000	82	2600 V3 for Video	801	VIS8400
16	S6900	84	2600F V3	802	VIS8600
17	S3900	85	5500F V3	803	VIS8800
18	S5500T	86	5600F V3	804	Cloud Protecto
19	S5600T	87	5800F V3	805	Dorado5000
20	S6800T	88	6800F V3	806	Dorado6000
21	S6900-M100	89	18500F V3	807	Dorado18000
22	S3900-M200	90	18800F V3		
23	S3900-M300	92	2800 V5		
24	S5900-M100	93	5300 V5		
25	S8100	94	5300F V5		
26	S5800T	95	5500 V5		
27	V1600N	96	5500F V5		
28	S5900-M200	97	5600 V5		
29	N8001	98	5600F V5		
30	S2600T	99	5800 V5		
31	S8000-I	100	5800F V5		
32	Dorado5100	101	6800 V5		
33	T3200	102	6800F V5		
34	T3500	103	18500 V5		
35	N8000-R710	104	18500F V5		
36	N8000-C2	105	18800 V5		
40	8000-C2-SYMC	106	18800F V5		

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
41	VTL	107	5500 V5 Elite		
42	S2900	108	2100 V3		
43	S2200T	112	2200V3 Enhanced		
44	Dorado2100_G2	113	2600V3 Enhanced		
45	VIS 6000	114	2600FV3 Enhanced		
46	HVS85T	115	5300V5 Enhanced		
47	HVS88T	116	5110 V5		
48	9000	117	5110F V5		
49	T3200 G2	118	5210 V5		
50	T3500 G2	119	5210F V5		
51	T3200 G3	120	5310 V5		
52	T3500 G3	121	5310F V5		
53	S5700	122	5510 V5		
54	CE6800	123	5510F V5		
55	CE12800	124	5610 V5		
56	18800F	125	5610F V5		
57	18500	126	5810 V5		
58	18800	127	5810F V5		
60	Dorado2100	128	6810 V5		
61	6800 V3	129	6810F V5		
62	6900 V3	130	18510 V5		

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